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Review

Aquaculture omics: An update on the current status of research and data analysis

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ABSTRACT

Aquaculture is the fast-growing agricultural sector and has the ability to meet the growing demand for protein nutritional security for future population. In future aquaculture is going to be the major source of fish proteins as capture fisheries reached at its maximum. However, several challenges need to overcome such as lack of genetically improved strains/varieties, lack of species-specific feed/functional feed, round the year availability of quality fish seed, pollution of ecosystems and increased frequencies of disease occurrence etc. In recent years, the continuous development of high throughput sequencing technology has revolutionized the biological sciences and provided necessary tools. Application of 'omics' in aquaculture research have been successfully used to resolve several productive and reproductive issues and thus ensure its sustainability and profitability. To date, high quality draft genomes of over fifty fish species have been generated and successfully used to develop large number of single nucleotide polymorphism markers (SNPs), marker panels and other genomic resources etc in several aquaculture species. Similarly, transcriptome profiling and miRNAs analysis have been used in aquaculture research to identify key transcripts and expression analysis of candidate genes/miRNAs involved in reproduction, immunity, growth, development, stress toxicology and disease. Metagenome analysis emerged as a promising scientific tool to analyze the complex genomes contained within microbial communities. Metagenomics has been successfully used in the aquaculture sector to identify novel and potential pathogens, antibiotic resistance genes, microbial roles in microcosms, microbial communities forming biofloc, probiotics etc. In the current review, we discussed application of high-throughput technologies (NGS) in the aquaculture sector.

1. Introduction

Fish is the cheapest source of quality animal protein and can meet the growing demand for animal protein. It is one of the sustainable sources of essential amino acids, proteins, vitamins and omega-3 fatty acids etc. Teleosts are the largest group of vertebrates comprising more than 32000 of species spreading across marine to freshwater habitat (Nelson et al., 2016), however till today only four hundred species are cultured (Gjedrem and Robinson, 2014). Currently, aquaculture is an emerging and increasingly growing agriculture sector (FAO, 2016). Globally, India ranks second for fisheries production, and China produces 1/3rd of the total fish harvested and 2/3rd of the fish cultivated (FAO, 2016). Though aquaculture has the potential to provide livelihood, rural employment and animal protein nutritional security it is not free from obstacles. One of the major obstacles of aquaculture growth is lack of availability of elite germplasm. Another challenging problem in aquaculture is fish diseases caused by several pathogens, such as bacteria, viruses, and fungi, showing high economic losses, and in turn, aquaculture results in environmental problems (Schwitzguébel and Wang, 2007). To make the aquaculture sector sustainable and profitable, it is essential to solve these problems using novel ideas and technologies.

During the last decade technological advancement accelerated fish genome sequencing enabling development of protocols/methods facilitate to study the global alteration of genes, proteins and metabolites expression level (Zampiga et al., 2018). The genome information of an organism remains unchanged during its entire period of life, however

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Table 1
Comparative account of NGS platforms.

Generation of sequencing technology	Sequencing Technology	NGS platform	Accuracy (%)	Read per Run (Bases)	GB per Run	Time per Run	Cost (USD)
2nd generation	Illumina	iSeq	>99.9%	2*150	0.3-1.2	1-10 days	~20,000
		MiniSeq	>99.9%	2*150	1.7-7.5	1-10 days	~50,000
		MiSeq	>99%	2*300	0.3-15	1-10 days	~100,000
		NextSeq	80%-99%	2*150	10-120	NA	250,000
		HiSeq	90%-99%	2150	10-1000	24hrs	650,000
	Thermo Fisher	PGM	87.5%	400	0.08-2.0	72 hrs	80,000
		S5	>99%	400	0.6-1.5	NA	60,000
		Proton	>99%	200	10-15	NA	149,000
3rd generation	Pacific Bioscience	PacBio RSII	90%	60,000	0.5-1.0	30 minutes to 2 hrs	750,000
		Sequel	99.9%	60,000	5-10	2 hrs	350,000
	Oxford Nanopore Technology	MinIon	99.3%	100,000+	10-20	6hrs	1000
		GridIon	99.3%	100,000+	50-100	72hrs	2400
		PromethIon	98%-99.1%	100,000+	480-960	72hrs	25,000
	Intelligent Biosystems (Qiagen)	2*100bp	NA	NA	Upto 80GB	40hrs	120,000

Table 2
Genome sequences of some aquaculture species

Species	Sequencing platform	Total size (Mb)	Contig N50 (Kb)	Scaffold N50 (Mb)	Percentages of genome	References
Stickleback	9.0X Sanger	463	83.2	10.8	86.9	(Jones et al., 2012)
Catfish	Illumina, PacBio	783	77.2	7.73	97.2	(Liu et al., 2016)
Grass carp	132X Illumina	900.5	40.8	6.46	64.0	(Wang et al., 2015)
Seabass	30X	675.4	53.2	5.09	86.0	(Tine et al., 2014)
Shark	454, Sanger	937	46.6	4.5	-	(Venkatesh et al., 2014)
Turbot	Illumina	544	31.2	4.3	-	(Figueras et al., 2016)
Tilapia	269X Illumina	1 010	29.3	2.80	70.9	(Brawand et al., 2014)
Cavefish	94X Illumina	964	14.7	1.78	-	(McGaugh et al., 2014)
Zebrafish	7.5X Sanger, Illumina	1 410	25.0	1.55	96.5	(Howe et al., 2013)
Medaka	10.6X Sanger	700.4	9.8	1.41	89.7	(Kasahara et al., 2007)
Platy fish	454, Illumina	669	22.0	1.1	90.2	(Schartl et al., 2013)
Common carp	454, Illumina, SOLiD	1690	68.4	1.0	51.8	(Xu et al., 2014a)
Tetraodon	8.3X Sanger	342.4	16.0	0.98	64.6	(Jaillon et al., 2004)
Coelacanth	Illumina	2 860	12.7	0.92	-	(Amemiya et al., 2013)
Sole	Illumina	477	26.5	0.87	93.3	(Chen et al., 2014)
Cod	454	753	7.1	0.69	44.1	(Star et al., 2011)
Yellow croacker	76X Illumina	644	25.7	0.50	-	(Wu et al., 2014)
Rainbow trout	70X Illumina	1 900	7.7	0.38	54. 0	(Berthelot et al., 2014)
Lamprey	Sanger	816	-	0.17	-	(Smith et al., 2013)
Fugu	5.6X Sanger	332.5	16.5	0.05-0.1	-	(Aparicio et al., 2002)
Rohu	Illumina, Ion torrent, Roche 454 and PacBio	1480	30.6	1.95	98	(Das et al., 2020)
Catla	Illumina and Nanopore	1010	-	0.7		(Sahoo et al., 2020)
Indian catfish	Illumina, Ion torrent, Roche 454, Nanopore and PacBio	941	-	1.3	94	(Kushwaha et al., 2021)

the resultant product of gene i.e the proteins or metabolites are subject to change at expression level in a rapid and dynamic way. Technological advancement allowed analysis of transcriptome data, and the resulting technology is known as RNA sequencing technology (RNA-seq technology) (Qian et al., 2014). RNA-seq technology overcomes many limitations of other transcriptomic methods, such as the microarray and tag-based sequencing method. Although RNA-seq methods have been available for a relatively short period, studies using the RNA-seq method have entirely reformed our perspective of the range and depth of eukaryotic transcriptomes. In applying the transcriptomics technology of the aquaculture sector, both model and non-model fish species have benefited from these sequencing methods and have undergone tremendous progress over the past several years. RNA-seq has helped to map and annotate fish transcriptomes, such as gene expression, gene regulatory networks, metabolic pathways, and protein-protein interaction networks, and has also provided an understanding of many biological processes in fish, such as adaptive evolution, development, stress response and host immune response (Qian et al., 2014). The functional classification of gene expression data is represented as matrices, and microarray data analysis mainly involves statistical analysis, classification, and clustering approaches (Can, 2014).

In this review, we first summarized the high-throughput sequencing technologies. We then reviewed the recent advances of whole-genome sequencing, transcriptomics, metagenome sequencing, miRNAs and single nucleotide polymorphisms (SNPs) and their applications in aquaculture sectors.

2. Whole-genome sequencing in aquaculture species

The availability of genome sequences is crucial for understanding a species' biology and serves as a roadmap to facilitate the development of technologies to achieve sustainable production. Aquaculture genomics started in the 1990s in the USA, Europe, China and other countries, although genome research started before the 1980s. The First

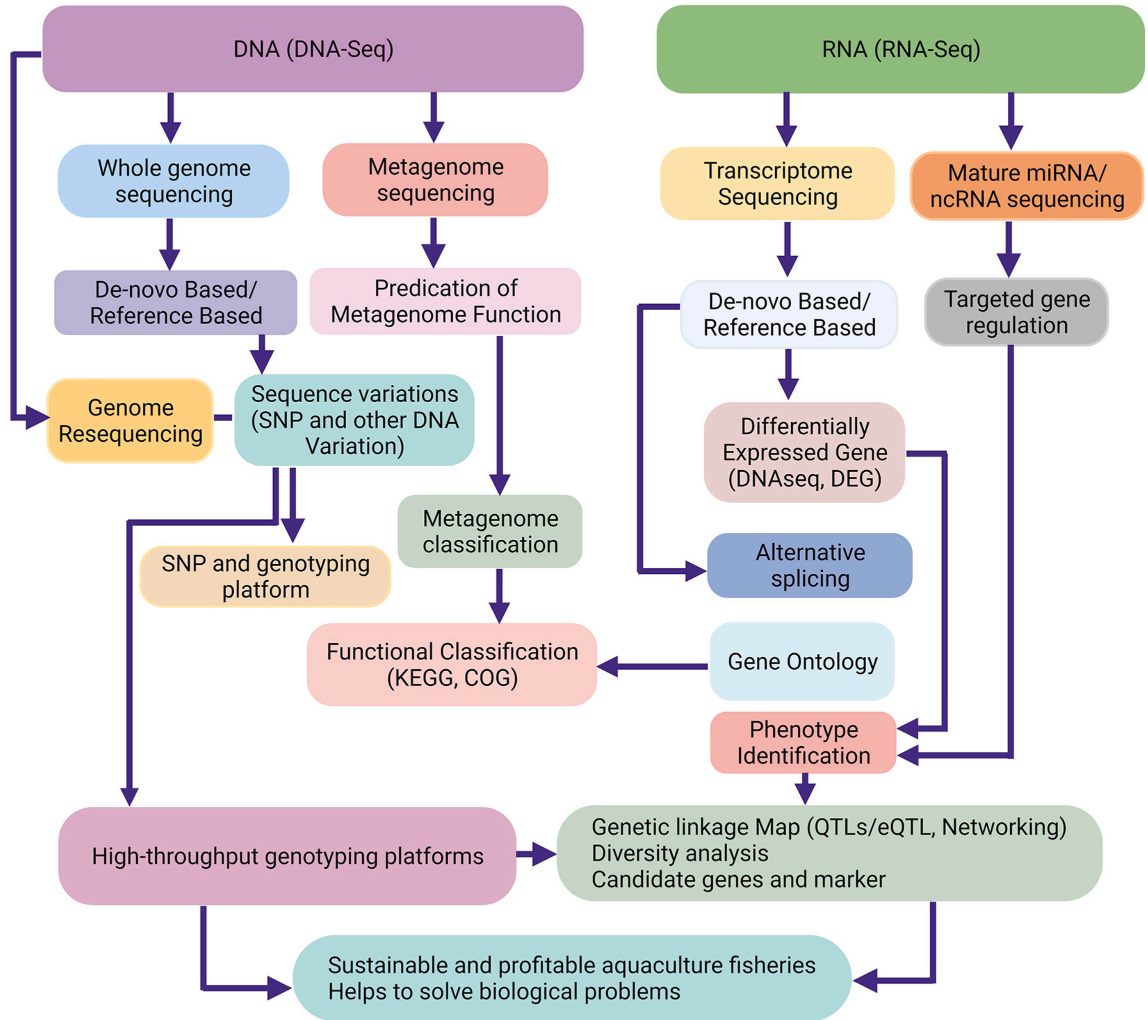

Fig. 1. Illustration of application of high throughput sequencing in fish genomics and genetics.

Aquaculture Genomics Workshop was held in 1977 at Dartmouth, Massachusetts, in the United States of America (https://www.globalseafood.org). For genome research, four species, catfish, salmonids, striped bass and tilapia, were the focus of the above mentioned genomics workshop. Through advanced sequencing technologies and the development of next-generation sequencing (NGS) platforms, rapid growth is presently made in the whole-genome sequencing of aquaculture species (Kumar and Kocour, 2017). To decipher the genome sequences of the aquaculture species, several sequencing technologies were used, such as Sanger sequencing, pyrosequencing (454 GS-FLX, Roche, Basel, Switzerland), sequencing by ligation (SOLiD, Life Technologies, California, USA), sequencing by synthesis (Illumina, California, USA), ion semiconductor (Ion Tor-rent), and single-molecule real-time sequencing (Pac-Bio). A comparative account of different NGS platforms with output parameters is provided in Table 1. Before 2005, the output of NGS platforms was less than one Gb/run; after 2005, the production of these platforms increased to ~1500 GB/run, and in 2016, the cost of sequencing was reduced 1000-fold/per base pair, and the bioinformatics analysis pipelines improved. Thus, the utility of whole-genome sequencing is ever-expanding in aquaculture species. In the recent past, whole-genome sequencing projects started with several different aquaculture species in various states of assembly. Currently, scientists are using single-molecule real-time (SMRT) sequencing, which means more short-read sequencing (SRS) in combination with very long-read sequencing (Vij et al., 2016; Roberts et al., 2013). The draft genome sequences of twenty-seven species, Pacific oyster, common carp, European seabass, grass carp, rohu, catla and Indian catfish, have been sequenced and published (Table 2).

The pyrosequencing was the first NGS technology available in the market and the Atlantic cod (*Gadus morhua*, cold-water marine fish) genome was sequenced employing this NGS platform only (shotgun and paired end library sequencing), and the genome size was estimated to be 830 Mb, with 13,000 contigs and more than 3800 scaffolds with N50 scaffold size of 488,312 Kb. Genome annotation identified 22,154 protein-coding genes (Star et al., 2011). The Pacific oyster (*Crassostrea gigas*) is an essential marine shellfish species having high percentages of repetitive sequences was sequenced using hybrid approach ie short read sequencing in combination with fosmid-pooling. The assembled genome was observed to be 559 Mb and 90 % of it was covered by the longest 1670 scaffolds with N50 size of 401 kb (Zhang et al., 2012). The whole-genome sequence of tetraploid common carp (*Cyprinus carpio*) has been completed in recent years (2014) by adopting multiplatform sequencing strategy ie Illumina, SOLiD and Roche 454 platform at the Chinese Academy of Fishery Sciences using both single-end and paired-end or mate-pair libraries of various insert size ranging from 250 bp to 8 kb and scaffold N50 length was observed to be 1.0 Mb (Xu et al., 2014a). The draft genome size is 1.69 Gb, contains 52,610 protein-coding genes, and 2503 scaffolds contributed to 90% of the assembly (Xu et al., 2014b). Another important freshwater species is Nile tilapia (*Oreochromis niloticus*), which has been cultured in more than 100 countries with over 6 million tons of annual production. Its genome was sequenced with four other species (*Neolamprologus brichardi*/pulcher, *Metriaclima zebra*,

Table 3
Transcriptome profiling in aquaculture species; purpose, targeted tissues and references

Transcriptome analysis in	Tissue	Species	References
Reproduction and development	Brain, Gonads, Liver, Pituitary	*Melanogrammusaeglefinus, Salmo salar, Gadusmorhua, Silurusmeridionalis, Silurusasotus, Labeorohita Gasterosteus aculeatus, Anguilla japonica, Oreochromis niloticus, Danio rerio, Gobiocyprisrarus*	(Das et al., 2020; Roy et al., 2018), (Sahu et al., 2015), (Kushwaha et al., 2021; Jin et al., 2015), (Avarre et al., 2014)
Immunity and disease	Spleen, Kidney, Brain, Muscle, Intestine, Heart, Skin	*Rainbow trout, Ctenopharyngodon Idella, Atlantic Salmon, Danio riror, Latescalcarifer, Nototheniacoriiceps, Brown trout, Salmo trutta*	(Chapman et al., 2014; Sudhagar et al., 2018) (Syahputra et al., 2019; Dang et al., 2016; Maekawa et al., 2019; Stockhammer et al., 2009; Kim et al., 2019)
Toxicology and stress	Embryos, Liver, Testis, Gills, Intestine, Swim bladder, Muscle	*Snout bream, Zebrafish, Larimichthyscrocea, Nile tilapia, Aristihthysnobilis, Cyprinus carpio, Gobicyprisrarus, Brown Trout, European flounder*	(Zheng et al., 2018; Asker et al., 2013), (Xu et al., 2015; Miao et al., 2018), (Wang et al., 2016; Zhang et al., 2017b)
Growth and nutrition	Brain, Pituitary, Liver, Muscle, Gut/ Intestine	*Rainbow trout, Atlantic salmon, Dicentrarchuslabrax, Danio rerio, Atlantic cod, Atlantic salmon, Sparus aurata, Aliivibriowodanis*	(Danzmann et al., 2016; Geay et al., 2011; De Santis et al., 2015), (Hjerde et al., 2015; Morrison and Wright, 2005)

Table 4
List of currently available gene prediction software packages

Sl. No.	Name	Links
1	AUGUSTUS	http://augustus.gobics.de/
2	CRITICA	http://www.ttaxus.com/software.html
3	ChemGenome	http://bgf.genomics.org.cn/
4	DNA SUBWAY	http://dnasubway.iplantcollaborative.org/
5	BGF	http://bgf.genomics.org.cn/
6	EUGENE	http://eugene.toulouse.inra.fr/
7	FGENESH	http://linux1.softberry.com/berry.phtml?topic=fgenesh&group=programs&subgroup=gfind
8	FRAMED	http://tata.toulouse.inra.fr/apps/FrameD/FD
9	GeneID	http://genome.crg.es/software/geneid/geneid.html
10	GeneMark	http://topaz.gatech.edu/GeneMark
11	GeneTack	http://topaz.gatech.edu/GeneTack

Pundamilia nyererei, and *Astatotilapia burtoni*) using sequencing by synthesis (SBS) technology. The assembled genome contained 77,578 contigs with a 29.5 kb N50 value and 13,517 scaffolds with a 2.8 Mb of N50 value. The genome length of the Nile tilapia (*Oreochromis niloticus*) is 815.7 Mb, containing 21,437 protein-coding genes, 22 pseudogenes, and 821 non-coding RNAs (Brawand et al., 2014). Additionally, whole-genome sequencing of goldfish was completed in 2019 using SMRT sequencing technology and found that the genome size was 1.6-2.08 pg. The assembled genome contained 9415 contigs with an N50 of 817 kbp and 1246 scaffolds (Chen et al., 2019). Similarly, the genome sequences of commercially important indigenous fish species such as *Labeo rohita*, *L. catla,* and *Clarias magur* have been sequenced (Das et al., 2020; Sahoo et al., 2020; Kushwaha et al., 2021). A multiplatform sequencing approach was adopted to sequence these genomes. Illumina short read and 454 pyrosequencing was used to decipher the *L. rohita* genome with scaffold N50 values of 1.95Mb and assembled genome size of 1.49 Gb. Similarly, the *L. catla* genome was sequenced using a combination of short read sequencing (Illumina) and long read sequencing (Oxford Nanopore) resulting an assembled size of 1.01 Gb with scaffold N50 value 0.7 Mb. Whereas a combination of Illumina, Iontorrent, 454 and Oxford Nanopre sequencing technology was used to decipher the genome sequence of *C. magur*. The assembled genome of C. magur was found to be 941 Mb with scaffold N50 values of 1.3 Mb and 23,748 estimated protein coding genes. The assembled genomes of other species, e.g., trobut, salmon, catfish and tilapia etc. are available in the NCBI SRA resources database. Still, the whole genome sequence of many aquaculture species is not accessible for public use. Hopefully, all reference genomes will be available in the public domain in the coming years. The availability of high-quality reference genomes will lead to an in-depth understanding of aquaculture species' production and reproductive biology, thereby enhancing aquaculture production.

Table 5
List of unique miRNAs identified and their functional role

Species	miRNAs	Function	References
Rainbow trout (*Oncorhynchus mykis*)	let-7, miR-10, miR-21, miR- 24, miR-25, miR-30, miR-143, miR-146, miR-148, and miR-202	Cell differentiation	(Ma et al., 2012)
Nile tilapia (*Oreochromis niloticus*)	let-7b, c,j, miR-1, miR-15a, miR-22a, miR-27a, miR-30b, miR-34, miR-125b, miR-133a,miR-140, miR-152, miR-192, miR-193a, miR-199, miR-204, miR-206, miR-214, miR-218a, b	Growth regulation	(Huang et al., 2012)
Zebra Fish (*Danio rerio*)	mir-430, mir-427, mir-520, mir-93, mir-20, mir-17	Regulates mRNAs overexpression	(Giraldez et al., 2006)
Teleost fish	mir-462, mir-731, mir-146, mir-21,mir-181, mir-155, mir-221, mir-1388, mir-100, mir-99	Immune response	(Andreassen and Høyheim, 2017)
Rohu (*Labeorohita)*	miR-22, miR-122, miR-365, miR-200, and miR-146, miR-216, miR-130, miR-456, miR-20, miR-23, miR-29, miR-200, miR-139, miR-1338, miR-199, miR-22, miR-100	carbohydrate metabolism	(Rasal et al., 2020)
Bata (*Labeobata*)	miR-202, miR-133, miR-132, and miR-29, miR-202, miR-132, and let-7, mir-132,mir-29	Liver specific	(Rasal et al., 2019)
Triploid Fish	miR-101a, miR-99, miR-101a, miR-100, miR-22a, miR-146a, miR-21, and miR-7a, mir-143	Testicular development	(Tao et al., 2018)
Zebra Fish (*Danio rerio*)	miR-148, mir-152, mir-144, mir-16c, mir-29b, mir-26, mir-133, mir-19, mir-152, mir-218, mir-101	Heart regeneration	(Klett et al., 2018)

Table 7
List of miRNA prediction tools/packages

Tool	Application	Description	Online Portal
MiRscan	Web server	A classic miRNA prediction server	http://genes.mit.edu/mirscan/
microPred	Packages	A tool to classify pre-miRNAs from pseudo hairpins and other ncRNAs	http://www.cs.ox.ac.uk/people/manohara.rukshan.batuwita/microPred.htm
miRAbela	Web server	A tool to predict if a sequence contains miRNA-like structures	http://www.mirz.unibas.ch/
miRNAFold	Web server	*Ab initio* miRNA prediction in genomes	http://evryrna.ibisc.univ-evry.fr/miRNAFold/
MirEval	Web server	A tool for evaluating sequences for possible miRNA-like properties	http://mimirna.centenary.org.au/mireval/
CID-miRNA	Web server and package	Identification of miRNAs from a single sequence or complete genome	https://github.com/alito/CIDmiRNA
miRanda	Web server	7 nt base pairing and weighted seed match, Energy calculation	http://cbio.mskcc.org/microrna_data/
RNAhybrid	Web service	Finds minimum free energy (MFE) of hybridization of long and short RNA	https://bibiserv2.cebitec.unibielefeld.de/rnahybrid
TargetScan	Web server	6–7 nucleotide (nt) base pairing, Z-score (Energy	http://www.targetscan.org/

3. Transcriptome study and data analysis in aquaculture species

Transcriptomics is the technique used to study an organism at the transcript level under particular conditions. Transcriptome data analysis gives a brief idea of molecular function and cellular processes that are active and dormant. Transcriptome analysis is used from a specific tissue (Li et al., 2014), a whole individual (Rodríguez-Celma et al., 2013), or a pool of individuals (Helm et al., 2013). The current progress in NGS technologies and advancement in computational tools have facilitated understanding the function/regulation of known/new genes at a genome-wide scale (Chaitankar et al., 2016). Transcriptome analysis workflow includes RNA isolation and purification methods involving TRIzol (phenol-chloroform), silica gel columns, and oligo dT-based approaches (Tan and Yiap, 2009). High-quality RNA with sufficient integrity (between 9 and 10) was taken for library preparation and sequencing purposes. Impure or degraded RNA will perform poorly in enzymatic applications. A graphical representation of the workflow for transcriptome data analysis is provided in Fig. 1. In the recent past, transcriptome analysis has been widely used in the aquaculture sector for effective identification and expression study of candidate/target genes involved in reproduction, growth, immunity, development, stress, disease, and toxicology (Garg et al., 2011). Recently, a project widely known as Fish-T1K was initiated to profile transcripts of 1000 fishes (Sun et al., 2016). Large numbers of transcripts, including novel transcripts, have been successfully identified in various organisms, including model and non-model aquaculture species, using high-throughput RNA sequencing technology (RNA-seq), such as channel catfish (*Ictalurus punctatus*) (Sun et al., 2016; Liu et al., 2016), zebrafish (*Danio rerio*) (Collins et al., 2012), European sea bass (*Dicentrarchus labrax*) (Sarropoulou et al., 2019), and rainbow trout *(Oncorhynchus mykiss)* (Palstra et al., 2013). In addition, liver-specific transcripts corresponding to various conditions have been identified in the liver (Liu et al., 2014; Qian et al., 2016; Zhang et al., 2017a; Yadetie et al., 2018; Dai et al., 2021).

Similarly, transcripts having a profound role in fish reproduction have been identified in several species (Reading et al., 2012; Cardoso et al., 2018; Yang et al., 2018; Tian et al., 2019; Saaristo et al., 2021). In addition, the regulation of glucose metabolism has been recently identified and demonstrated to have similar regulation patterns and activity in model zebrafish and mammals (Zhang et al., 2018). Although the application of transcriptome analysis in the aquaculture sector is challenging, the literature search at PubMed/Google with the keyword fish and RNA-seq indicates that the publication percentages in the aquaculture field (particularly in fish) have increased considerably in the past 3-4 years. Some target tissues for transcriptome sequencing analysis of various fish are listed in Table 3.

High-throughput technology (RNA-seq) is also used to identify novel transcript regions in the genome. RNA-seq results found a large number of unknown transcribed regions in all fish species, including rohu carp (Robinson et al., 2012; Sahu et al., 2015), zebrafish (Pauli et al., 2012) and rainbow trout (Palstra et al., 2013). Among them, many ncRNAs are essential due to their crucial roles in various biological functions (BFs) and cellular processes (CCs), such as gene regulation, genome defense, translation and RNA splicing (Mattick and Makunin, 2006). For example, more than 500 unique long-strand ncRNAs were found in zebrafish embryos (Ulitsky et al., 2011). In addition, 19,015 unique genes were observed in the ovarian tissue of crucian carp (Wang et al., 2019), 940 reproduction-related genes involving 184 reproduction, 223 hormone-activity, and receptorbinding genes, 178 receptor-activity genes, and 355 embryonic-development-related proteins were identified in *Labeo rohita* (Hamilton) (Sahu et al., 2015). Nearly 809 reproduction-related genes were identified in *Cyprinus carpio* (Anitha et al., 2019). These newly identified transcripts will help further

Table 6
Metagenome information of some aquaculture species

Species	Tissue	Host microbes	Genus	OTUs	References
Sea Cucumber	Intestine	Bacteroidetes, Rhodobacterales, Flavobacteriaceae,	*Vibro, Rhodobacteraceae,*	>90%	(Zhang et al., 2019)
Freshwater Fish	Intestine	Fusobacteria	*Serratia, Bacteroides, Cetobacterium, Clostridium, Bifidobacterium, Lactobacillus, Streptococcus*	94%	(Tsuchiya et al., 2008; Tang et al., 2019)
Common carps, Asian carp, Prussian carp	Gut and Intestine	Proteobacteria, Firmicutes, Fusobacteria and Cyanobacteria	*Cetobacterium, Aeromonas, Acinetobacter, Clostridium, Streptococcus, Streptophyte*	95%	(Ni et al., 2014; Ye et al., 2014), (Li et al., 2015; Kashinskaya et al., 2015)
Nile tilapia	Gut	Proteobacteria Fusobacteria, Firmicutes, Actinobacteria, Bacteroidets	*Cetobacterium, Aquaspirillum, Edwardsiella* and *Plesiomonas*	92%	(Zhang et al., 2017c; Adeoye et al., 2016)
European eels	Skin mucus	Gamma proteobacteria, Flavobacteria, Betaproteobacteria, Alphaproteobacteria	*Vibrio, Pseudomonas, Stenotrophomonas, Achromobactin, Agrobacterium, Xanthomonas*	30 -95%	(Carda Diéguez et al., 2014)
Common crap, Atlantic salmon	Skin–mucus	Proteobacteria, Fusobacteria	*Lysobacter, Amylibacter, Methylobacterium, Burkholderia, Ratstonia*	96%	(Li et al., 2013; Llewellyn et al., 2017)
Thresher shark	Skin–mucus	Proteobacteria, Bacteroidets, Cyanobacteria	Pseudoalteromonas, Erythrobaacteria, Marinobacteria,*Psudomonas,Sphingopyxi, Alteromonas*	91-79%	(Doane et al., 2017)

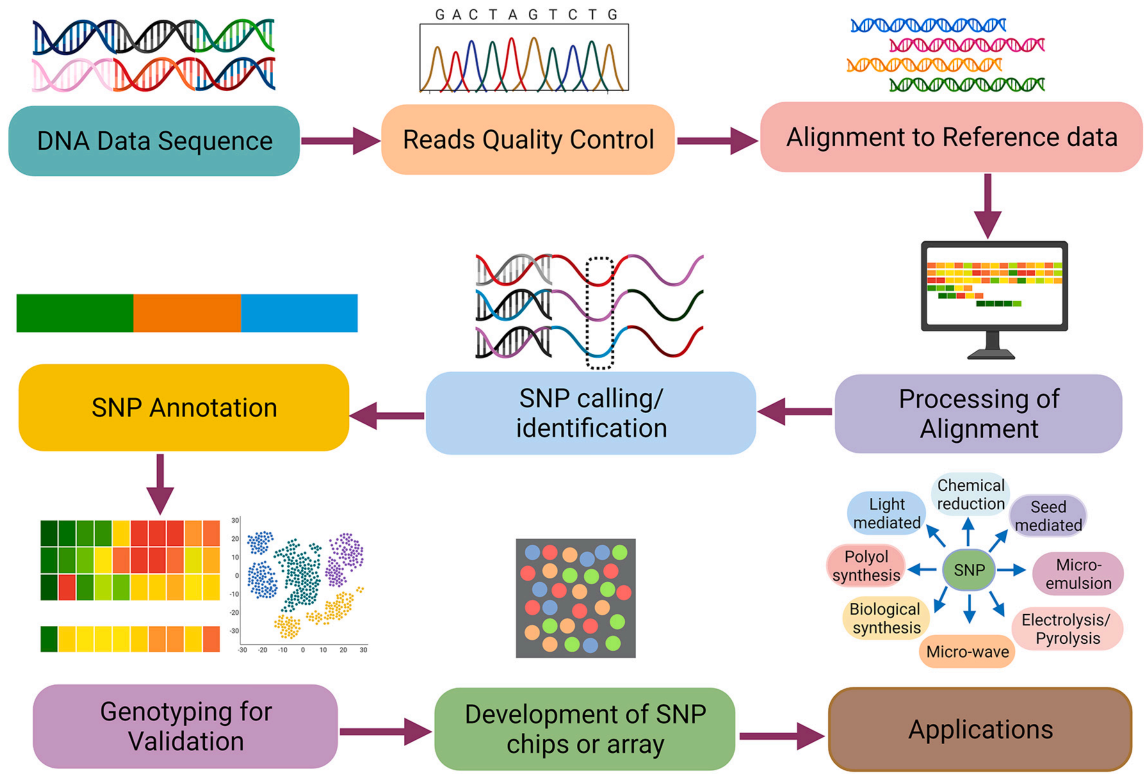

Fig. 2. Steps for SNP mining.

annotation and functional analysis of the sequenced genome.

4. MicroRNA discovery and data analysis in aquaculture species

MicroRNAs (miRNAs) are small noncoding endogenous single-stranded RNAs of 19–25 nucleotides in length that play an essential role in developmental biology by gene modulation via post-transcriptional gene regulation during acclimatization and adaptation in the case of adverse conditions or climate change (Rasal et al., 2016). MicroRNAs are present in the 3' untranslated region (UTR) of the gene (Kaeuferle et al., 2014) and have substantial contributions to regulatory networks of development and adaptive plasticity in fishes (Rasal et al., 2016). High-throughput sequencing technology has been widely used to discover miRNAs in several organisms (Salem et al., 2010; Zhu and Leung, 2015). Changes in miRNA expression in animals have been shown during larval and juvenile growth (Campos et al., 2014), during larval ontogeny (Bizuayehu et al., 2015), in eggs (Ma et al., 2012), and in skin pigmentation (Yan et al., 2013). The miRNA data were generated/discovered using marker discovery, gene discovery, variant identification, and miRNA identification processes. miRNA function was predicted by using bioinformatics tools. Some target gene prediction tools and their online web portals are shown in Table 4. Huang et al. (Huang et al., 2015) studied miRNAs in 11 different fishes and predicted 43 miRNAs belonging to 38 miRNA families with their target genes through expressed sequence tags (EST) and genome sequence surveys (GSSs). Then, some selected miRNAs were validated through real-time PCR. These findings suggested that computational methods are more convenient for screening miRNAs and their target genes/proteins/pathways from non-model fishes. Juanchich et al. (Juanchich et al., 2016) found 2946 miRNAs in 16 different tissues of rainbow trout using high-throughput sequencing technology. We have presented information about some individual miRNAs and their functions in Table 5. Recently, mature miRNAs of some fish species, such as *Danio rerio* (346), *Salmo salar* (371), *Takifugu rubripes* (175), *Cyprinus carpio* (134), *Tetraodon nigroviridis* (132), *Oryzias latipes* (168), *Ictalurus punctatus* (281), *Paralichthys olivaceus* (20), and *Hippoglossus hippoglossus* (40) (Hsu et al., 2014), have been documented.

Kloosterman and coworkers identified 66 novel and 139 known miRNAs from 5-day-old *Danio rerio* larvae. They suggested that out of 205 miRNAs, miR-153 regulates snap25 during synaptic transmission and motor neuron development and that miR-27 targets ptk2.2 to regulate pharyngeal arch morphogenesis in zebrafish (Kloosterman et al., 2007). Tozzini and coworkers studied an annual fish (*Nothobranchius furzeri*) and identified 165 conserved miRNAs in the brain, whereas 17–92 microRNAs were mainly associated with brain neurogenesis (Terzibasi Tozzini et al., 2014). Recently, 138 conserved and 161 novel miRNAs were identified in the liver tissue of *Labeo rohita* and *Labeo bata* and revealed that miR-200, miR-22, miR-365, miR-122, and miR-146 are mainly involved in the carbohydrate metabolism pathway (Rasal et al., 2019; Rasal et al., 2020). Evidence is increasing daily in support of the critical role of miRNAs regulating various biological processes in aquaculture species.

5. Metagenome profiling and data analysis in aquaculture species

Microbes play an essential role in various ecosystems; however, not all the microbes present in an environment have been characterized. The metagenomics approach provides detailed information about the microbiome structure present in a particular environment (Simon and Daniel, 2011; Larsen et al., 2014). It is commonly used in microbial ecology to study microbial communities in more detail, including many microbes/strains that cannot be cultivated in the laboratory. Previously, scientists used traditional disease control methods such as antibiotics and vaccines to control the pathogen. However, rising public pressures against antibiotics in animals and aquaculture shifted the focus from killing pathogens to promoting beneficial microbes. Therefore, studies on the effects of prebiotics on fish and associated microbiomes have been published in the last decade. Recently, sequencing and bioinformatics analyses have made it possible to mine massive metagenomic

Table 8
SNPs identified in aquatic animals.

Species	Sequencing Platform used	Nos of SNP discovered	Computational tools used	Validation methodology	References
Chinook salmon (*Oncorhynchus tshawytscha*)	Sanger sequencing	40	tagRepeats.perl	Taqman	(Smith et al., 2005)
Chum salmon (*O. keta*)	Sanger sequencing	155	tagRepeats.perl	Taqman	(Smith et al., 2005)
Sockeye salmon (*O.nerka*)	Sanger sequencing	114	tagRepeats.perl	Taqman	(Smith et al., 2005)
Atlantic salmon (*Salmo salar*)	Sanger sequencing	19	Phred/Phrap/ PolyPhred series of base-calling	Sanger sequencing	(Ryynanen and Primmer, 2006)
Brown trout (*Salmo trutta*)	Sanger sequencing	15	Phred/Phrap/ PolyPhred series of base-calling	Sanger sequencing	(Ryynanen and Primmer, 2006)
Arctic char (*Salvelinusalpinus*)	Sanger sequencing	2	Phred/Phrap/ PolyPhred series of base-calling	Sanger Sequencing	(Ryynanen and Primmer, 2006)
Atlantic cod (*Gadusmorhua*)	Sanger sequencing	724	Phred/Phrap/ PolyBayes series of base-calling	MassARRAY Sequenome	(Moen et al., 2008)
Rainbow trout (*O.mykiss*)	454	13140–24 627	ssahaSNP program	Golden Gate Assay	(Sanchez et al., 2009)
Lake whitefish (*Coregonus spp.*)	454	6042	ssahaSNP program	SNPplex	(Renaut et al., 2010)
Rainbow trout (*O. Mykiss*)	454 GSFLX titanium	58000	ssahaSNP program	Golden Gate Assay	(Sanchez et al., 2011)
Common carp (*C. carpio*)	Illumina Hiseq2000	712,042 Intra-strain SNPs (483,276 SNPs mirror carp, 486,629 SNPs purse red carp, 478,028 SNPs for Xingguo red carp and 488,281 SNPs for Yellow River carp	BWA and SAMtools	Sanger sequencing and PCR amplification	(Xu et al., 2012)
European anchovy (*E.encrasicolus*)	454 and Illumina	2317	SAMtools	Taqman genotyping assay	(Montes et al., 2013)
Indo-pacific dolphin	Illumina	3681	SOAPsnp	NA	(Gui et al., 2013)
Pufferfish (*Takifugurubripes*)	Illumina	62270	BWA and SAMtools	PCR amplification and direct sequencing	(Cui et al., 2014)
Bay scallop (*Argopectenirradians*)	Illumina	32,206 and 23,312 SNPs for northern and southern bay scallops.	SOAPsnp		(Du et al., 2014)
Channel Catfish (*I. punctatus*)	Illumina	407,861	SAMtools	Not done	(Sun et al., 2014)
Half-smooth Tongue sole (*C. semilaevis*)	454	21234	ssahaSNP program	Not done	(Wang et al., 2014)
White Shrimp (*L. Vannamei*)	Illumina Hiseq2000	96040	BWA and SAMtools	Sanger sequencing and PCR amplification	(Yu et al., 2014)
Olive Flounder (*P. olivaceus*)	EST seq	514 SNPs	EST-EST alignment	Taqman	(Kim et al., 2014)
Zebrafish (*D. rerio*)	Illumina	164	CLCBio workbench	SequenomMassARRAY	(Ulloa et al., 2015)
Pacu (*Piaractusmesopotamicus*)	Illumina	32	BWA and SAMtools	PCR	(Mastrochirico-Filho et al., 2016)
Rainbow trout (Oncorhynchus mykiss)	Illumina	59,112, 87,066	SAMtools and GATK tool	Sanger sequencing and PCR amplification	(Al-Tobasei et al., 2017)
southern catfish (*Silurus meridionalis*)	Restriction site-associated DNA (RAD) sequencing	26714	SOAPsnp tool	PCR	(Xie et al., 2018)
Tilapia species (*Oreochromis niloticus*, *O. aureus*, *O. mossambicus, O. u. hornorum*)	Double digested restriction-site associated DNA (ddRAD) sequencing	57 species-specific SNP markers	Stacks v1.46	PCR-based KASP assays	(Syaifudin et al., 2019)
Pacific oyster (*Crassostrea gigas*)	RAD	21,499	Xiom Analysis Suite (version 3.1, Affymetrix), FreeMapTools	PCR	(Vendrami et al., 2019)
Large yellow croaker (*Larimichthys crocea*)	double-digest restriction-site associated DNA (ddRAD)	5,261 SNP	JoinMap4.1	PCR amplification and direct sequencing	(Kong et al., 2019)

datasets and discover general patterns that govern microbial ecosystems. High-throughput sequencing technology has been widely used for making thousands of microbial genome information at a low cost. These data have been produced using two approaches: shotgun and (16S ribosomal RNA [rRNA] gene) amplicon metagenomic sequencing methods, which identify functional genes at a large scale. For analyzing and visualizing the amplicon and metagenomic data, several tools/online servers have been used, such as IMG/M (Chen et al., 2017), MG-RAST (Keegan et al., 2016), EBI Metagenomics (Hunter et al., 2013), and some computational pipelines, such as MEGAN (Huson et al., 2007), Mothur (Neves et al., 2017), and QIIME (Siegwald et al., 2017), without command-line operations or solid computational knowledge (Table 7). Metagenome sequencing has been documented in some fish species, such as zebrafish (Semova et al., 2012), Atlantic salmon (Zarkasi et al., 2016), Seabream (Kormas et al., 2014), rainbow trout (Wong et al., 2013), grass carp (Ni et al., 2014), Antarctic peninsula (Cowart et al., 2018), Malaysian mahseer (Tan et al., 2019) and Arctic charr (Nyman et al., 2017). Bacterial community profiles in the fish microbiota were observed in response to various factors affecting the host, including variations in temperature, stress, salinity, digestive physiology, developmental stage, climate change, and feeding strategy. Information regarding the fish species and their common microbes is shown in Table 6. However, metagenomics has also shed light on community function when backed up with metatranscriptomic data (RNA-seq of environmental samples), providing information regarding essential microbial survival in a particular condition. This dual approach has revealed that some groups of archaea, generally known as ammonia oxidizers, have the physiological machinery to digest proteins, carbohydrates, and lipids (Li et al., 2015), prompting a complete rethinking of the carbon cycle in aquaculture environments.

6. Current status of single nucleotide polymorphism (SNP) analysis in aquaculture species

Single nucleotide polymorphism (SNP) markers are considered decisive genetic markers. They signify the most acceptable resolve possible for a marker map (at the single nucleotide level), are codominant, biallelic and commonly plentiful in populations with a low mutation rate. However, the DNA sequencing approach is the gold standard of SNP detection. Various techniques/tools have been used for SNP discovery. Scanning for changes in targeted sites in the genome was the primary effort to identify SNPs arbitrarily in the human genome (Botstein et al., 1980). However, genotyping methods involve complex processes, specific apparatuses, expensive reagents, and require a high skill level (Watanabe et al., 2002).

Moreover, with the invention of PCR, DNA sequencing (Andreassen et al., 2010) and oligonucleotide synthesis were coming into existence but were costly until quite recently. However, several high-throughput SNP genotyping techniques, such as Sequenom MassArray (Gabriel et al., 2009), Illumina Bead Array (Oliphant et al., 2002), and Affymetrix Axiom Array (Rabbee and Speed, 2006), have recently become available. The critical step for SNP mining is given in Fig. 2. Much research has been carried out on SNPs in aquatic animals in the last decade. Baetscher et al. (Baetscher et al., 2017) developed and reported two sets of 96 SNP assays for two species of anadromous alosine fishes, alewife and blueback herring, and successfully resolved the stock structure of these two species of North America (Baetscher et al., 2017). In tilapia (family Cichlidae), species-specific markers were analyzed to distinguish ten tilapia species (Syaifudin et al., 2019). Using double digested restriction-site associated DNA (ddRAD) sequencing to discover SNP markers, a total of 57 species-specific SNP markers originated between the samples analyzed from the ten tilapia species (Syaifudin et al., 2019). Similarly, ~2 million SNPs were identified in the Indian major carp, *L. rohita*. Here, we have comprehensively presented NGS and other platforms with computational tools for SNP discovery in aquaculture species, as given in Table 8.

7. Conclusion and future outlook

In conclusion, the application of NGS technology and computational tools/software toward the sustainable enhancement of aquaculture production has been underused compared to its use in other livestock sectors. An increasing number of genome sequences of commercially important aquaculture species need to be available to accelerate the development of genomics resources, such as marker databases, marker panels, marker maps, and QTLs, among others, to implement MAS and GS toward rapid genetic improvement and elite germplasm development. Furthermore, the availability of genome sequences will accelerate the in-depth understanding of the molecular mechanisms underlying production and reproductive traits. Literature on nutrigenomics studies in aquaculture species is scanty in comparison to other livestock species. Further its application to develop species-specific and functional feeds needs to be explored in commercially important aquaculture species. Transcriptome profiling and differential gene expression analysis technology must be employed to identify further vital transcripts/genes involved in growth, disease resistance, and immunity. The identification of miRNAs, essential production and reproductive trait regulators, is underused in aquaculture species. Metagenomics has great potential to identify key microbial pathways and metabolites responsible for important physiological and biological processes in aquaculture species. Furthermore, it has enormous potential to determine the importance of novel species-specific probiotics of aquaculture. In summary, application of NGS technology and other techniques/tools will significantly impact aquaculture production in the coming decades.

Declaration of Competing Interest

The authors have no conflict of interest.

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References

Adeoye, A.A., Jaramillo-Torres, A., Fox, S.W., Merrifield, D.L., Davies, S.J., 2016. Supplementation of formulated diets for tilapia (Oreochromis niloticus) with selected exogenous enzymes: Overall performance and effects on intestinal histology and microbiota. Anim. Feed Sci. Technol. 215, 133–143.

Al-Tobasei, R., Ali, A., Leeds, T.D., Liu, S., Palti, Y., Kenney, B., Salem, M., 2017. Identification of SNPs associated with muscle yield and quality traits using allelic-imbalance analyses of pooled RNA-Seq samples in rainbow trout. BMC Genomics 18, 582.

Amemiya, C.T., Alföldi, J., Lee, A.P., Fan, S., Philippe, H., MacCallum, I., Braasch, I., Manousaki, T., Schneider, I., Rohner, N., Organ, C., Chalopin, D., Smith, J.J., Robinson, M., Dorrington, R.A., Gerdol, M., Aken, B., Biscotti, M.A., Barucca, M., Baurain, D., Berlin, A.M., Blatch, G.L., Buonocore, F., Burmester, T., Campbell, M.S., Canapa, A., Cannon, J.P., Christoffels, A., De Moro, G., Edkins, A.L., Fan, L., Fausto, A.M., Feiner, N., Forconi, M., Gamieldien, J., Gnerre, S., Gnirke, A., Goldstone, J.V., Haerty, W., Hahn, M.E., Hesse, U., Hoffmann, S., Johnson, J., Karchner, S.I., Kuraku, S., Lara, M., Levin, J.Z., Litman, G.W., Mauceli, E., Miyake, T., Mueller, M.G., Nelson, D.R., Nitsche, A., Olmo, E., Ota, T., Pallavicini, A., Panji, S., Picone, B., Ponting, C.P., Prohaska, S.J., Przybylski, D., Saha, N.R., Ravi, V., Ribeiro, F.J., Sauka-Spengler, T., Scapigliati, G., Searle, S.M.J., Sharpe, T., Simakov, O., Stadler, P.F., Stegeman, J.J., Sumiyama, K., Tabbaa, D., Tafer, H., Turner-Maier, J., van Heusden, P., White, S., Williams, L., Yandell, M., Brinkmann, H., Volff, J.-N., Tabin, C.J., Shubin, N., Schartl, M., Jaffe, D.B., Postlethwait, J.H., Venkatesh, B., Di Palma, F., Lander, E.S., Meyer, A., Lindblad-Toh, K., 2013. The African coelacanth genome provides insights into tetrapod evolution. Nature 496, 311–316.

Andreassen, R., Høyheim, B., 2017. miRNAs associated with immune response in teleost fish. Dev. Comp. Immunol. 75, 77–85.

Andreassen, R., Lunner, S., Hoyheim, B., 2010. Targeted SNP discovery in Atlantic salmon (Salmo salar) genes using a 3'UTR-primed SNP detection approach. BMC Genomics 11, 706.

Anitha, A., Gupta, Y.-R., Deepa, S., Ningappa, M., Rajanna, K.B., Senthilkumaran, B., 2019. Gonadal transcriptome analysis of the common carp, Cyprinus carpio: Identification of differentially expressed genes and SSRs. Gen. Comp. Endocrinol. 279, 67–77.

Aparicio, S., Chapman, J., Stupka, E., Putnam, N., Chia, J.-M., Dehal, P., Christoffels, A., Rash, S., Hoon, S., Smit, A., Gelpke, M.D.S., Roach, J., Oh, T., Ho, I.Y., Wong, M., Detter, C., Verhoef, F., Predki, P., Tay, A., Lucas, S., Richardson, P., Smith, S.F., Clark, M.S., Edwards, Y.J.K., Doggett, N., Zharkikh, A., Tavtigian, S.V., Pruss, D., Barnstead, M., Evans, C., Baden, H., Powell, J., Glusman, G., Rowen, L., Hood, L., Tan, Y.H., Elgar, G., Hawkins, T., Venkatesh, B., Rokhsar, D., Brenner, S., 2002. Whole-genome shotgun assembly and analysis of the genome of Fugu rubripes. Science 297, 1301–1310.

Asker, N., Kristiansson, E., Albertsson, E., Larsson, D.G.J., Förlin, L., 2013. Hepatic transcriptome profiling indicates differential mRNA expression of apoptosis and immune related genes in eelpout (Zoarces viviparus) caught at Göteborg harbor, Sweden. Aquat. Toxicol. 130-131, 58–67.

Avarre, J.C., Guinand, B., Dugue, R., Cosson, J., Legendre, M., Panfili, J., Durand, J.D., 2014. Plasticity of gene expression according to salinity in the testis of broodstock and F1 black-chinned tilapia, Sarotherodon melanotheron heudelotii. PeerJ 2, e702.

Baetscher, D.S., Hasselman, D.J., Reid, K., Palkovacs, E.P., Garza, J.C., 2017. Discovery and characterization of single nucleotide polymorphisms in two anadromous alosine fishes of conservation concern. Ecol. Evol. 7, 6638–6648.
Berthelot, C., Brunet, F., Chalopin, D., Juanchich, A., Bernard, M., Noël, B., Bento, P., Da Silva, C., Labadie, K., Alberti, A., Aury, J.-M., Louis, A., Dehais, P., Bardou, P., Montfort, J., Klopp, C., Cabau, C., Gaspin, C., Thorgaard, G.H., Boussaha, M., Quillet, E., Guyomard, R., Galiana, D., Bobe, J., Volff, J.-N., Genêt, C., Wincker, P., Jaillon, O., Roest Crollius, H., Guiguen, Y., 2014. The rainbow trout genome provides novel insights into evolution after whole-genome duplication in vertebrates. Nat. Commun. 5, 3657.
Bizuayehu, T., Johansen, S., Puvanendran, V., Toften, H., Babiak, I., 2015. Temperature during early development has long-term efects on microRNA expression in Atlantic cod. BMC Genomics 16, 305.
Botstein, D., White, R.L., Skolnick, M., Davis, R.W., 1980. Construction of a genetic linkage map in man using restriction fragment length polymorphisms. Am. J. Hum. Genet. 32, 314–331.
Brawand, D., Wagner, C.E., Li, Y.I., Malinsky, M., Keller, I., Fan, S., Simakov, O., Ng, A. Y., Lim, Z.W., Bezault, E., Turner-Maier, J., Johnson, J., Alcazar, R., Noh, H.J., Russell, P., Aken, B., Alföldi, J., Amemiya, C., Azzouzi, N., Baroiller, J.-F., Barloy-Hubler, F., Berlin, A., Bloomquist, R., Carleton, K.L., Conte, M.A., D'Cotta, H., Eshel, O., Gaffney, L., Galibert, F., Gante, H.F., Gnerre, S., Greuter, L., Guyon, R., Haddad, N.S., Haerty, W., Harris, R.M., Hofmann, H.A., Hourlier, T., Hulata, G., Jaffe, D.B., Lara, M., Lee, A.P., MacCallum, I., Mwaiko, S., Nikaido, M., Nishihara, H., Ozouf-Costaz, C., Penman, D.J., Przybylski, D., Rakotomanga, M., Renn, S.C.P., Ribeiro, F.J., Ron, M., Salzburger, W., Sanchez-Pulido, L., Santos, M.E., Searle, S., Sharpe, T., Swofford, R., Tan, F.J., Williams, L., Young, S., Yin, S., Okada, N., Kocher, T.D., Miska, E.A., Lander, E.S., Venkatesh, B., Fernald, R.D., Meyer, A., Ponting, C.P., Streelman, J.T., Lindblad-Toh, K., Seehausen, O., Di Palma, F., 2014. The genomic substrate for adaptive radiation in African cichlid fish. Nature 513, 375–381.
Campos, C., Sundaram, A.Y.M., Valente, L.M.P., Conceição, L.E.C., Engrola, S., Fernandes, J.M.O., 2014. Thermal plasticity of the miRNA transcriptome during Senegalese sole development. BMC Genomics 15, 525.
Can, T., 2014. Introduction to Bioinformatics. In: Yousef, M., Allmer, J. (Eds.), miRNomics: MicroRNA Biology and Computational Analysis Springer. Humana Press, Totowa, NJ.
Carda Diéguez, M., Ghai, R., Rodriguez-Valera, F., Amaro, C., 2014. Metagenomics of the mucosal microbiota of European Eels. Genome Announc. 2 (1132-1114).
Cardoso, S.D., Goncalves, D., Goesmann, A., Canario, A.V.M., Oliveira, R.F., 2018. Temporal variation in brain transcriptome is associated with the expression of female mimicry as a sequential male alternative reproductive tactic in fish. Mol. Ecol. 27, 789–803.
Chaitankar, V., Karakülah, G., Ratnapriya, R., Giuste, F.O., Brooks, M.J., Swaroop, A., 2016. Next generation sequencing technology and genomewide data analysis: Perspectives for retinal research. Prog. Retin. Eye Res. 55, 1–31.
Chapman, J.M., Proulx, C.L., Veilleux, M.A., Levert, C., Bliss, S., Andre, M.E., Lapointe, N.W., Cooke, S.J., 2014. Clear as mud: a meta-analysis on the effects of sedimentation on freshwater fish and the effectiveness of sediment-control measures. Water Res. 56, 190–202.
Chen, S., Zhang, G., Shao, C., Huang, Q., Liu, G., Zhang, P., Song, W., An, N., Chalopin, D., Volff, J.-N., Hong, Y., Li, Q., Sha, Z., Zhou, H., Xie, M., Yu, Q., Liu, Y., Xiang, H., Wang, N., Wu, K., Yang, C., Zhou, Q., Liao, X., Yang, L., Hu, Q., Zhang, J., Meng, L., Jin, L., Tian, Y., Lian, J., Yang, J., Miao, G., Liu, S., Liang, Z., Yan, F., Li, Y., Sun, B., Zhang, H., Zhang, J., Zhu, Y., Du, M., Zhao, Y., Schartl, M., Tang, Q., Wang, J., 2014. Whole-genome sequence of a flatfish provides insights into ZW sex chromosome evolution and adaptation to a benthic lifestyle. Nat. Genet. 46, 253–260.
Chen, I.M.A., Markowitz, V.M., Chu, K., Palaniappan, K., Szeto, E., Pillay, M., Ratner, A., Huang, J., Andersen, E., Huntemann, M., Varghese, N., Hadjithomas, M., Tennessen, K., Nielsen, T., Ivanova, N.N., Kyrpides, N.C., 2017. IMG/M: integrated genome and metagenome comparative data analysis system. Nucleic Acids Res. 45, D507–D516.
Chen, Z., Omori, Y., Koren, S., Shirokiya, T., Kuroda, T., Miyamoto, A., Wada, H., Fujiyama, A., Toyoda, A., Zhang, S., Wolfsberg, T.G., Kawakami, K., Phillippy, A.M., Mullikin, J.C., Burgess, S.M., 2019. De novo assembly of the goldfish (Carassius auratus) genome and the evolution of genes after whole-genome duplication. Sci. Adv. 5, eaav0547.
Collins, J.E., White, S., Searle, S.M.J., Stemple, D.L., 2012. Incorporating RNA-seq data into the zebrafish Ensembl genebuild. Genome Res. 22, 2067–2078.
Cowart, D.A., Murphy, K.R., Cheng, C.H.C., 2018. Metagenomic sequencing of environmental DNA reveals marine faunal assemblages from the West Antarctic Peninsula. Mar. Genomics 37, 148–160.
Cui, J., Wang, H., Liu, S., Zhu, L., Qiu, X., Jiang, Z., Wang, X., Liu, Z., 2014. SNP discovery from transcriptome of the swimbladder of Takifugu rubripes. PLoS One 9, e92502.
Dai, Y.F., Shen, Y.B., Wang, S.T., Zhang, J.H., Su, Y.H., Bao, S.C., Xu, X.Y., Li, J.L., 2021. RNA-Seq transcriptome analysis of the liver and brain of the Black Carp (Mylopharyngodon piceus) during fasting. Mar. Biotechnol. 23 (3), 389–401.
Dang, Y., Xu, X., Shen, Y., Hu, M., Zhang, M., Li, L., Lv, L., Li, J., 2016. Transcriptome analysis of the innate immunity-related complement system in spleen tissue of ctenopharyngodon idella infected with aeromonas hydrophila. PLoS One 11, e0157413.
Danzmann, R.G., Kocmarek, A.L., Norman, J.D., Rexroad, C.E., Palti, Y., 2016. Transcriptome profiling in fast versus slow-growing rainbow trout across seasonal gradients. BMC Genomics 17, 60.
Das, P., Sahoo, L., Das, S.P., Bit, A., Joshi, C.G., Kushwaha, B., Kumar, D., Shah, T.M., Hinsu, A.T., Patel, N., Patnaik, S., Agarwal, S., Pandey, M., Srivastava, S., Meher, P. K., Jayasankar, P., Koringa, P.G., Nagpure, N.S., Kumar, R., Singh, M., Iquebal, M.A., Jaiswal, S., Kumar, N., Raza, M., Das Mahapatra, K., Jena, J., 2020. De novo assembly and genome-wide SNP discovery in Rohu Carp, Labeo rohita. Front. Genet. 11, 386.
De Santis, C., Taylor, J.F., Martinez-Rubio, L., Boltana, S., Tocher, D.R., 2015. Influence of development and dietary phospholipid content and composition on intestinal transcriptome of Atlantic Salmon (Salmo salar). PLoS One 10, e0140964.
Doane, M.P., Haggerty, J.M., Kacev, D., Papudeshi, B., Dinsdale, E.A., 2017. The skin microbiome of the common thresher shark (Alopias vulpinus) has low taxonomic and gene function β-diversity. Environ. Microbiol. Rep. 9, 357–373.
Du, X., Li, L., Zhang, S., Meng, F., Zhang, G., 2014. SNP identification by transcriptome sequencing and candidate gene-based association analysis for heat tolerance in the bay scallop Argopecten irradians. PLoS One 9, e104960.
FAO, 2016. Fisheries and Aquaculture Department The State ofWorld Fisheries and Aquaculture. Food and Agriculture Organization of the United Nations.
Figueras, A., Robledo, D., Corvelo, A., Hermida, M., Pereiro, P., Rubiolo, J.A., Gómez-Garrido, J., Carreté, L., Bello, X., Gut, M., Gut, I.G., Marcet-Houben, M., Forn-Cuní, G., Galán, B., García, J.L., Abal-Fabeiro, J.L., Pardo, B.G., Taboada, X., Fernández, C., Vlasova, A., Hermoso-Pulido, A., Guigó, R., Álvarez-Dios, J.A., Gómez-Tato, A., Viñas, A., Maside, X., Gabaldón, T., Novoa, B., Bouza, C., Alioto, T., Martínez, P., 2016. Whole genome sequencing of turbot (Scophthalmus maximus; Pleuronectiformes): a fish adapted to demersal life. DNA Res. 23, 181–192.
Gabriel, S., Ziaugra, L., Tabbaa, D., 2009. SNP genotyping using the Sequenom Massarray iPLEX platform, Current protocols in human genetics / editorial board, Jonathan L. Haines ... [et al.], Chapter 2 (Unit 2.12).
Garg, R., Patel, R.K., Jhanwar, S., Priya, P., Bhattacharjee, A., Yadav, G., Bhatia, S., Chattopadhyay, D., Tyagi, A.K., Jain, M., 2011. Gene discovery and tissue-specific transcriptome analysis in chickpea with massively parallel pyrosequencing and web resource development. Plant Physiol. 156, 1661–1678.
Geay, F., Ferraresso, S., Zambonino-Infante, J.L., Bargelloni, L., Quentel, C., Vandeputte, M., Kaushik, S., Cahu, C.L., Mazurais, D., 2011. Effects of the total replacement of fish-based diet with plant-based diet on the hepatic transcriptome of two European sea bass (Dicentrarchus labrax) half-sibfamilies showing different growth rates with the plant-based diet. BMC Genomics 12, 522.
Giraldez, A.J., Mishima, Y., Rihel, J., Grocock, R.J., Van Dongen, S., Inoue, K., Enright, A. J., Schier, A.F., 2006. Zebrafish MiR-430 promotes deadenylation and clearance of maternal mRNAs. Science 312, 75–79.
Gjedrem, T., Robinson, N., 2014. Advances by selective breeding for aquatic species: a review. Agric. Sci. 05, 1152–1158.
Gui, D., Jia, K., Xia, J., Yang, L., Chen, J., Wu, Y., Yi, M., 2013. De novo assembly of the Indo-Pacific humpback dolphin leucocyte transcriptome to identify putative genes involved in the aquatic adaptation and immune response. PLoS One 8, e72417.
Helm, R.R., Siebert, S., Tulin, S., Smith, J., Dunn, C.W., 2013. Characterization of differential transcript abundance through time during Nematostella vectensis development. BMC Genomics 14, 266.
Hjerde, E., Karlsen, C., Sørum, H., Parkhill, J., Willassen, N.P., Thomson, N.R., 2015. Co-cultivation and transcriptome sequencing of two co-existing fish pathogens Moritella viscosa and Aliivibrio wodanis. BMC Genomics 16, 447.
Howe, K., Clark, M.D., Torroja, C.F., Torrance, J., Berthelot, C., Muffato, M., Collins, J.E., Humphray, S., McLaren, K., Matthews, L., McLaren, S., Sealy, I., Caccamo, M., Churcher, C., Scott, C., Barrett, J.C., Koch, R., Rauch, G.-J., White, S., Chow, W., Kilian, B., Quintais, L.T., Guerra-Assunção, J.A., Zhou, Y., Gu, Y., Yen, J., Vogel, J.-H., Eyre, T., Redmond, S., Banerjee, R., Chi, J., Fu, B., Langley, E., Maguire, S.F., Laird, G.K., Lloyd, D., Kenyon, E., Donaldson, S., Sehra, H., Almeida-King, J., Loveland, J., Trevanion, S., Jones, M., Quail, M., Willey, D., Hunt, A., Burton, J., Sims, S., McLay, K., Plumb, B., Davis, J., Clee, C., Oliver, K., Clark, R., Riddle, C., Elliot, D., Threadgold, G., Harden, G., Ware, D., Begum, S., Mortimore, B., Kerry, G., Heath, P., Phillimore, B., Tracey, A., Corby, N., Dunn, M., Johnson, C., Wood, J., Clark, S., Pelan, S., Griffiths, G., Smith, M., Glithero, R., Howden, P., Barker, N., Lloyd, C., Stevens, C., Harley, J., Holt, K., Panagiotidis, G., Lovell, J., Beasley, H., Henderson, C., Gordon, D., Auger, K., Wright, D., Collins, J., Raisen, C., Dyer, L., Leung, K., Robertson, L., Ambridge, K., Leongamornlert, D., McGuire, S., Gilderthorp, R., Griffiths, C., Manthravadi, D., Nichol, S., Barker, G., Whitehead, S., Kay, M., Brown, J., Murnane, C., Gray, E., Humphries, M., Sycamore, N., Barker, D., Saunders, D., Wallis, J., Babbage, A., Hammond, S., Mashreghi-Mohammadi, M., Barr, L., Martin, S., Wray, P., Ellington, A., Matthews, N., Ellwood, M., Woodmansey, R., Clark, G., Cooper, J.D., Tromans, A., Grafham, D., Skuce, C., Pandian, R., Andrews, R., Harrison, E., Kimberley, A., Garnett, J., Fosker, N., Hall, R., Garner, P., Kelly, D., Bird, C., Palmer, S., Gehring, I., Berger, A., Dooley, C. M., Ersan-Ürün, Z., Eser, C., Geiger, H., Geisler, M., Karotki, L., Kirn, A., Konantz, J., Konantz, M., Oberländer, M., Rudolph-Geiger, S., Teucke, M., Lanz, C., Raddatz, G., Osoegawa, K., Zhu, B., Rapp, A., Widaa, S., Langford, C., Yang, F., Schuster, S.C., Carter, N.P., Harrow, J., Ning, Z., Herrero, J., Searle, S.M.J., Enright, A., Geisler, R., Plasterk, R.H.A., Lee, C., Westerfield, M., de Jong, P.J., Zon, L.I., Postlethwait, J.H., Nüsslein-Volhard, C., Hubbard, T.J.P., Roest Crollius, H., Rogers, J., Stemple, D.L., 2013. The zebrafish reference genome sequence and its relationship to the human genome. Nature 496, 498–503.
Hsu, A., Chen, S.-J., Chang, Y.-S., Chen, H.-C., Chu, P.-H., 2014. Systemic approach to identify serum microRNAs as potential biomarkers for acute myocardial infarction. Biomed. Res. Int. 2014, 418628.
Huang, C., Li, Y.-H., Hu, S.-Y., Chi, J., Lin, G., Lin, C., Gong, H.-Y., Chen, R., Chang, S., Liu, F., Wu, J., 2012. Differential expression patterns of growth-related microRNAs in the skeletal muscle of Nile tilapia. J. Anim. Sci. 90.

Huang, Y., Zou, Q., Ren, H.T., Sun, X.H., 2015. Prediction and characterization of microRNAs from eleven fish species by computational methods, Saudi. J. Biol. Sci. 22, 374–381.
Hunter, S., Corbett, M., Denise, H., Fraser, M., Gonzalez-Beltran, A., Hunter, C., Jones, P., Leinonen, R., McAnulla, C., Maguire, E., Maslen, J., Mitchell, A., Nuka, G., Oisel, A., Pesseat, S., Radhakrishnan, R., Rocca-Serra, P., Scheremetjew, M., Sterk, P., Vaughan, D., Cochrane, G., Field, D., Sansone, S.-A., 2013. EBI metagenomics—a new resource for the analysis and archiving of metagenomic data. Nucleic Acids Res. 42, D600–D606.
Huson, D.H., Auch, A.F., Qi, J., Schuster, S.C., 2007. MEGAN analysis of metagenomic data. Genome Res. 17, 377–386.
Jaillon, O., Aury, J.-M., Brunet, F., Petit, J.-L., Stange-Thomann, N., Mauceli, E., Bouneau, L., Fischer, C., Ozouf-Costaz, C., Bernot, A., Nicaud, S., Jaffe, D., Fisher, S., Lutfalla, G., Dossat, C., Segurens, B., Dasilva, C., Salanoubat, M., Levy, M., Boudet, N., Castellano, S., Anthouard, V., Jubin, C., Castelli, V., Katinka, M., Vacherie, B., Biémont, C., Skalli, Z., Cattolico, L., Poulain, J., De Berardinis, V., Cruaud, C., Duprat, S., Brottier, P., Coutanceau, J.-P., Gouzy, J., Parra, G., Lardier, G., Chapple, C., McKernan, K.J., McEwan, P., Bosak, S., Kellis, M., Volff, J.-N., Guigó, R., Zody, M.C., Mesirov, J., Lindblad-Toh, K., Birren, B., Nusbaum, C., Kahn, D., Robinson-Rechavi, M., Laudet, V., Schachter, V., Quétier, F., Saurin, W., Scarpelli, C., Wincker, P., Lander, E.S., Weissenbach, J., Roest Crollius, H., 2004. Genome duplication in the teleost fish Tetraodon nigroviridis reveals the early vertebrate proto-karyotype. Nature 431, 946–957.
Jin, F.F., Boucharel, J., Lin, I.I., et al., 2015. Reply. Nature 526, E5–E6.
Jones, F.C., Grabherr, M.G., Chan, Y.F., Russell, P., Mauceli, E., Johnson, J., Swofford, R., Pirun, M., Zody, M.C., White, S., Birney, E., Searle, S., Schmutz, J., Grimwood, J., Dickson, M.C., Myers, R.M., Miller, C.T., Summers, B.R., Knecht, A.K., Brady, S.D., Zhang, H., Pollen, A.A., Howes, T., Amemiya, C., Baldwin, J., Bloom, T., Jaffe, D.B., Nicol, R., Wilkinson, J., Lander, E.S., Di Palma, F., Lindblad-Toh, K., Kingsley, D.M., P., 2012. Broad Institute Genome Sequencing, T. Whole Genome Assembly, The genomic basis of adaptive evolution in threespine sticklebacks. Nature 484, 55–61.
Juanchich, A., Bardou, P., Rué, O., Gabillard, J.-C., Gaspin, C., Bobe, J., Guiguen, Y., 2016. Characterization of an extensive rainbow trout miRNA transcriptome by next generation sequencing. BMC Genomics 17, 164.
Kaeuferle, T., Bartel, S., Dehmel, S., Krauss-Etschmann, S., 2014. MicroRNA methodology: advances in miRNA technologies. Methods Mol. Biol. 1169, 121–130.
Kasahara, M., Naruse, K., Sasaki, S., Nakatani, Y., Qu, W., Ahsan, B., Yamada, T., Nagayasu, Y., Doi, K., Kasai, Y., Jindo, T., Kobayashi, D., Shimada, A., Toyoda, A., Kuroki, Y., Fujiyama, A., Sasaki, T., Shimizu, A., Asakawa, S., Shimizu, N., Hashimoto, S.-I., Yang, J., Lee, Y., Matsushima, K., Sugano, S., Sakaizumi, M., Narita, T., Ohishi, K., Haga, S., Ohta, F., Nomoto, H., Nogata, K., Morishita, T., Endo, T., Shin-I, T., Takeda, H., Morishita, S., Kohara, Y., 2007. The medaka draft genome and insights into vertebrate genome evolution. Nature 447, 714–719.
Kashinskaya, E.N., Belkova, N.L., Izvekova, G.I., Simonov, E.P., Andree, K.B., Glupov, V. V., Baturina, O.A., Kabilov, M.R., Solovyev, M.M., 2015. A comparative study on microbiota from the intestine of Prussian carp (Carassius gibelio) and their aquatic environmental compartments, using different molecular methods. J. Appl. Microbiol. 119, 948–961.
Keegan, K.P., Glass, E.M., Meyer, F., 2016. MG-RAST, a metagenomics service for analysis of microbial community structure and function. Methods Mol. Biol. 1399, 207–233.
Kim, J.E., Lee, Y.M., Lee, J.H., Noh, J.K., Kim, H.C., Park, C.J., Park, J.W., Kim, K.K., 2014. Development and Validation of Single Nucleotide Polymorphism (SNP) Markers from an Expressed Sequence Tag (EST) Database in Olive Flounder (Paralichthys olivaceus). Balsaenggwa Saengsig 18, 275–286.
Kim, Y.C., Kwon, W.J., Min, J.G., Kim, K.I., Jeong, H.D., 2019. Complete genome sequence and pathogenic analysis of a new betanodavirus isolated from shellfish. J. Fish Dis. 42, 519–531.
Klett, H., Jürgensen, L., Most, P., Busch, M., Günther, F., Dobreva, G., Leuschner, F., Hassel, D., Busch, H., Boerries, M., 2018. Delineating the dynamic transcriptome response of mRNA and microRNA during Zebrafish heart regeneration. Biomolecules 9, 11.
Kloosterman, W.P., Lagendijk, A.K., Ketting, R.F., Moulton, J.D., Plasterk, R.H.A., 2007. Targeted Inhibition of miRNA Maturation with Morpholinos Reveals a Role for miR-375 in Pancreatic Islet Development. PLoS Biol. 5, e203.
Kong, S., Ke, Q., Chen, L., Zhou, Z., Pu, F., Zhao, J., Bai, H., Peng, W., Xu, P., 2019. Constructing a High-Density Genetic Linkage Map for Large Yellow Croaker (Larimichthys crocea) and Mapping Resistance Trait Against Ciliate Parasite Cryptocaryon irritans. Mar. Biotechnol. 21.
Kormas, K.A., Meziti, A., Mente, E., Frentzos, A., 2014. Dietary differences are reflected on the gut prokaryotic community structure of wild and commercially reared sea bream (Sparus aurata). MicrobiologyOpen 3, 718–728.
Kumar, G., Kocour, M., 2017. Applications of next-generation sequencing in fisheries research: A review. Fish. Res. 186, 11–22.
Kushwaha, B., Pandey, M., Das, P., Joshi, C.G., Nagpure, N.S., Kumar, R., Kumar, D., Agarwal, S., Srivastava, S., Singh, M., Sahoo, L., Jayasankar, P., Meher, P.K., Shah, T. M., Hinsu, A.T., Patel, N., Koringa, P.G., Das, S.P., Patnaik, S., Bit, A., Iquebal, M.A., Jaiswal, S., Jena, J., 2021. The genome of walking catfish Clarias magur (Hamilton, 1822) unveils the genetic basis that may have facilitated the development of environmental and terrestrial adaptation systems in air-breathing catfishes. DNA Res. 28.
Larsen, A.M., Mohammed, H.H., Arias, C.R., 2014. Characterization of the gut microbiota of three commercially valuable warmwater fish species. J. Appl. Microbiol. 116, 1396–1404.
Li, X., Yan, Q., Xie, S., Hu, W., Yu, Y., Hu, Z., 2013. Gut microbiota contributes to the growth of fast-growing transgenic common carp (Cyprinus carpio L.). PLoS One 8, e64577.
Li, B., Tsoi, L.C., Swindell, W.R., Gudjonsson, J.E., Tejasvi, T., Johnston, A., Ding, J., Stuart, P.E., Xing, X., Kochkodan, J.J., Voorhees, J.J., Kang, H.M., Nair, R.P., Abecasis, G.R., Elder, J.T., 2014. Transcriptome analysis of psoriasis in a large case-control sample: RNA-seq provides insights into disease mechanisms. J. Invest. Dermatol. 134, 1828–1838.
Li, H., Limenitakis, J.P., Fuhrer, T., Geuking, M.B., Lawson, M.A., Wyss, M., Brugiroux, S., Keller, I., Macpherson, J.A., Rupp, S., Stolp, B., Stein, J.V., Stecher, B., Sauer, U., McCoy, K.D., Macpherson, A.J., 2015. The outer mucus layer hosts a distinct intestinal microbial niche. Nat. Commun. 6, 8292.
Liu, S., Gao, G., Palti, Y., Cleveland, B.M., Weber, G.M., Rexroad III, C.E., 2014. RNA-seq analysis of early hepatic response to handling and confinement stress in rainbow trout. PLoS One 9, e88492.
Liu, Z., Liu, S., Yao, J., Bao, L., Zhang, J., Li, Y., Jiang, C., Sun, L., Wang, R., Zhang, Y., Zhou, T., Zeng, Q., Fu, Q., Gao, S., Li, N., Koren, S., Jiang, Y., Zimin, A., Xu, P., Phillippy, A.M., Geng, X., Song, L., Sun, F., Li, C., Wang, X., Chen, A., Jin, Y., Yuan, Z., Yang, Y., Tan, S., Peatman, E., Lu, J., Qin, Z., Dunham, R., Li, Z., Sonstegard, T., Feng, J., Danzmann, R.G., Schroeder, S., Scheffler, B., Duke, M.V., Ballard, L., Kucuktas, H., Kaltenboeck, L., Liu, H., Armbruster, J., Xie, Y., Kirby, M.L., Tian, Y., Flanagan, M.E., Mu, W., Waldbieser, G.C., 2016. The channel catfish genome sequence provides insights into the evolution of scale formation in teleosts. Nat. Commun. 7 (11757-11757).
Llewellyn, M.S., Leadbeater, S., Garcia, C., Sylvain, F.E., Custodio, M., Ang, K.P., Powell, F., Carvalho, G.R., Creer, S., Elliot, J., Derome, N., 2017. Parasitism perturbs the mucosal microbiome of Atlantic Salmon. Sci. Rep. 7, 43465.
Ma, H., Hostuttler, M., Wei, H., Rexroad III, C.E., Yao, J., 2012. Characterization of the rainbow trout egg MicroRNA transcriptome. PLoS One 7, e39649.
Maekawa, S., Wang, P.-C., Chen, S.-C., 2019. Comparative study of immune reaction against bacterial infection from transcriptome analysis. Front. Immunol. 10, 153.
Mastrochirico-Filho, V.A., Hata, M.E., Sato, L.S., Jorge, P.H., Foresti, F., Rodriguez, M.V., Martínez, P., Porto-Foresti, F., Hashimoto, D.T., 2016. SNP discovery from liver transcriptome in the fish Piaractus mesopotamicus. Conserv. Genet. Resour. 8, 109–114.
Mattick, J.S., Makunin, I.V., 2006. Non-coding RNA. Hum. Mol. Genet. 15, R17–R29.
McGaugh, S.E., Gross, J.B., Aken, B., Blin, M., Borowsky, R., Chalopin, D., Hinaux, H., Jeffery, W.R., Keene, A., Ma, L., Minx, P., Murphy, D., O'Quin, K.E., Rétaux, S., Rohner, N., Searle, S.M.J., Stahl, B.A., Tabin, C., Volff, J.-N., Yoshizawa, M., Warren, W.C., 2014. The cavefish genome reveals candidate genes for eye loss. Nat. Commun. 5, 5307.
Miao, S., Zhao, C., Zhu, J., Hu, J., Dong, X., Sun, L., 2018. Dietary soybean meal affects intestinal homoeostasis by altering the microbiota, morphology and inflammatory cytokine gene expression in northern snakehead. Sci. Rep. 8, 113.
Moen, T., Hayes, B., Nilsen, F., Delghandi, M., Fjalestad, K.T., Fevolden, S.E., Berg, P.R., Lien, S., 2008. Identification and characterisation of novel SNP markers in Atlantic cod: evidence for directional selection. BMC Genet. 9, 18.
Montes, I., Conklin, D., Albaina, A., Creer, S., Carvalho, G.R., Santos, M., Estonba, A., 2013. SNP discovery in European anchovy (Engraulis encrasicolus, L) by high-throughput transcriptome and genome sequencing. PLoS One 8, e70051.
Morrison, C., Wright, J., 2005. A study of the histology of the digestive tract of the Nile tilapia. J. Fish Biol. 54, 597–606.
Nelson, S.J., Grande, C.T., Wilson, V.H.M., 2016. Fishes of the World, 5th ed., p. 752
Neves, A.L.A., Li, F., Ghoshal, B., McAllister, T., Guan, L.L., 2017. Enhancing the resolution of rumen microbial classification from metatranscriptomic data using kraken and mothur. Front. Microbiol. 8, 2445.
Ni, J., Yan, Q., Yu, Y., Zhang, T., 2014. Factors influencing the grass carp gut microbiome and its effect on metabolism. FEMS Microbiol. Ecol. 87, 704–714.
Nyman, A., Huyben, D., Lundh, T., Dicksved, J., 2017. Effects of microbe- and mussel-based diets on the gut microbiota in Arctic charr (Salvelinus alpinus). Aquacult. Rep. 5, 34–40.
Oliphant, A., Barker, D.L., Stuelpnagel, J.R., Chee, M.S., 2002. BeadArray technology: enabling an accurate, cost-effective approach to high-throughput genotyping. BioTechniques 56-58. Suppl. (60-51).
Palstra, A.P., Beltran, S., Burgerhout, E., Brittijn, S.A., Magnoni, L.J., Henkel, C.V., Jansen, H.J., van den Thillart, G.E.E.J.M., Spaink, H.P., Planas, J.V., 2013. Deep RNA Sequencing of the Skeletal Muscle Transcriptome in Swimming Fish. PLoS One 8, e53171.
Pauli, A., Valen, E., Lin, M.F., Garber, M., Vastenhouw, N.L., Levin, J.Z., Fan, L., Sandelin, A., Rinn, J.L., Regev, A., Schier, A.F., 2012. Systematic identification of long noncoding RNAs expressed during zebrafish embryogenesis. Genome Res. 22, 577–591.
Qian, X., Ba, Y., Zhuang, Q., Zhong, G., 2014. RNA-Seq technology and its application in fish transcriptomics. Omics J. Integr. Biol. 18, 98–110.
Qian, B., Xue, L., Huang, H., 2016. Liver transcriptome analysis of the large yellow croaker (Larimichthys crocea) during fasting by using RNA-Seq. PLoS One 11, e0150240.
Rabbee, N., Speed, T.P., 2006. A genotype calling algorithm for affymetrix SNP arrays. Bioinformatics 22, 7–12.
Rasal, K.D., Chakrapani, V., Patra, S.K., Mohapatra, S.D., Nayak, S., Jena, S., Sundaray, J. K., Jayasankar, P., Barman, H.K., 2016. Identification of deleterious mutations in myostatin gene of rohu Carp (Labeo rohita) using modeling and molecular dynamic simulation approaches. Biomed. Res. Int. 2016, 7562368.
Rasal, K.D., Iquebal, M.A., Jaiswal, S., Dixit, S., Vasam, M., Nandi, S., Raza, M., Sahoo, L., Angadi, U.B., Rai, A., Kumar, D., Sundaray, J.K., 2019. liver-specific microrna

identification in farmed carp, labeo bata (hamilton, 1822), fed with starch diet using high-throughput sequencing. Mar. Biotechnol. 21, 589–595.
Rasal, K.D., Iquebal, M.A., Pandey, A., Behera, P., Jaiswal, S., Vasam, M., Dixit, S., Raza, M., Sahoo, L., Nandi, S., Angadi, U.B., Rai, A., Kumar, D., Nagpure, N., Chaudhari, A., Sundaray, J.K., 2020. Revealing liver specific microRNAs linked with carbohydrate metabolism of farmed carp, Labeo rohita (Hamilton, 1822). Genomics 112, 32–44.
Reading, B.J., Chapman, R.W., Schaff, J.E., Scholl, E.H., Opperman, C.H., Sullivan, C.V., 2012. An ovary transcriptome for all maturational stages of the striped bass (Morone saxatilis), a highly advanced perciform fish. BMC Res. Notes 5, 111.
Renaut, S., Nolte, A.W., Bernatchez, L., 2010. Mining transcriptome sequences towards identifying adaptive single nucleotide polymorphisms in lake whitefish species pairs (Coregonus spp. Salmonidae). Mol. Ecol. 19 (Suppl. 1), 115–131.
Roberts, R.J., Carneiro, M.O., Schatz, M.C., 2013. The advantages of SMRT sequencing. Genome Biol. 14, 405.
Robinson, N., Sahoo, P.K., Baranski, M., Mahapatra, K.D., Saha, J.N., Das, S., Mishra, Y., Das, P., Barman, H.K., Eknath, A.E., 2012. Expressed sequences and polymorphisms in Rohu Carp (Labeo rohita, Hamilton) revealed by mRNA-seq. Mar. Biotechnol. 14, 620–633.
Rodríguez-Celma, J., Pan, I.C., Li, W., Lan, P., Buckhout, T.J., Schmidt, W., 2013. The transcriptional response of Arabidopsis leaves to Fe deficiency, Front. Plant Sci. 4, 276.
Roy, S., Saha, T.T., Zou, Z., Raikhel, A.S., 2018. Regulatory pathways controlling female insect reproduction. Annu. Rev. Entomol. 63, 489–511.
Ryynanen, H.J., Primmer, C.R., 2006. Single nucleotide polymorphism (SNP) discovery in duplicated genomes: intron-primed exon-crossing (IPEC) as a strategy for avoiding amplification of duplicated loci in Atlantic salmon (Salmo salar) and other salmonid fishes. BMC Genomics 7, 192.
Saaristo, M., Craft, J.A., Tyagi, S., Johnstone, C.P., Allinson, M., Ibrahim, K.S., Wong, B. B.M., 2021. Transcriptome-wide changes associated with the reproductive behaviour of male guppies exposed to 17alpha-ethinyl estradiol. Environ. Pollut. 270, 116286.
Sahoo, L., Das, P., Sahoo, B., Das, G., Meher, P.K., Udit, U.K., Mahapatra, K.D., Sundaray, J.K., 2020. The draft genome of Labeo catla. BMC Res. Notes 13, 411.
Sahu, D.K., Panda, S.P., Meher, P.K., Das, P., Routray, P., Sundaray, J.K., Jayasankar, P., Nandi, S., 2015. Construction, De-Novo Assembly and Analysis of Transcriptome for Identification of Reproduction-Related Genes and Pathways from Rohu, Labeo rohita (Hamilton). PLoS One 10, e0132450.
Salem, M., Xiao, C., Womack, J., Rexroad, C.E., Yao, J., 2010. A MicroRNA repertoire for functional genome research in rainbow trout (Oncorhynchus mykiss). Mar. Biotechnol. 12, 410–429.
Sanchez, C.C., Smith, T.P., Wiedmann, R.T., Vallejo, R.L., Salem, M., Yao, J., Rexroad 3rd, C.E., 2009. Single nucleotide polymorphism discovery in rainbow trout by deep sequencing of a reduced representation library. BMC Genomics 10, 559.
Sanchez, C.C., Weber, G.M., Gao, G., Cleveland, B.M., Yao, J., Rexroad 3rd, C.E., 2011. Generation of a reference transcriptome for evaluating rainbow trout responses to various stressors. BMC Genomics 12, 626.
Sarropoulou, E., Kaitetzidou, E., Papandroulakis, N., Tsalafouta, A., Pavlidis, M., 2019. Inventory of European Sea Bass (Dicentrarchus labrax) sncRNAs vital during early teleost development. Front. Genet. 10, 657.
Schartl, M., Walter, R.B., Shen, Y., Garcia, T., Catchen, J., Amores, A., Braasch, I., Chalopin, D., Volff, J.-N., Lesch, K.-P., Bisazza, A., Minx, P., Hillier, L., Wilson, R.K., Fuerstenberg, S., Boore, J., Searle, S., Postlethwait, J.H., Warren, W.C., 2013. The genome of the platyfish, Xiphophorus maculatus, provides insights into evolutionary adaptation and several complex traits. Nat. Genet. 45, 567–572.
Schwitzguébel, J.-P., Wang, H., 2007. Environmental impact of aquaculture and countermeasures to aquaculture pollution in China. Environ. Sci. Pollut. Res. 14, 452–462.
Semova, I., Carten, Juliana D., Stombaugh, J., Mackey, Lantz C., Knight, R., Farber, Steven A., Rawls, John F., 2012. Microbiota regulate intestinal absorption and metabolism of fatty acids in the zebrafish. Cell Host Microbe 12, 277–288.
Siegwald, L., Touzet, H., Lemoine, Y., Hot, D., Audebert, C., Caboche, S., 2017. Assessment of common and emerging bioinformatics pipelines for targeted metagenomics. PLoS One 12, e0169563.
Simon, C., Daniel, R., 2011. Metagenomic analyses: past and future trends. Appl. Environ. Microbiol. 77, 1153–1161.
Smith, C.T., Elfstrom, C.M., Seeb, L.W., Seeb, J.E., 2005. Use of sequence data from rainbow trout and Atlantic salmon for SNP detection in Pacific salmon. Mol. Ecol. 14, 4193–4203.
Smith, J.J., Kuraku, S., Holt, C., Sauka-Spengler, T., Jiang, N., Campbell, M.S., Yandell, M.D., Manousaki, T., Meyer, A., Bloom, O.E., Morgan, J.R., Buxbaum, J.D., Sachidanandam, R., Sims, C., Garruss, A.S., Cook, M., Krumlauf, R., Wiedemann, L. M., Sower, S.A., Decatur, W.A., Hall, J.A., Amemiya, C.T., Saha, N.R., Buckley, K.M., Rast, J.P., Das, S., Hirano, M., McCurley, N., Guo, P., Rohner, N., Tabin, C.J., Piccinelli, P., Elgar, G., Ruffier, M., Aken, B.L., Searle, S.M.J., Muffato, M., Pignatelli, M., Herrero, J., Jones, M., Brown, C.T., Chung-Davidson, Y.-W., Nanlohy, K.G., Libants, S.V., Yeh, C.-Y., McCauley, D.W., Langeland, J.A., Pancer, Z., Fritzsch, B., de Jong, P.J., Zhu, B., Fulton, L.L., Theising, B., Flicek, P., Bronner, M.E., Warren, W.C., Clifton, S.W., Wilson, R.K., Li, W., 2013. Sequencing of the sea lamprey (Petromyzon marinus) genome provides insights into vertebrate evolution. Nat. Genet. 45, 415–421.
Star, B., Nederbragt, A.J., Jentoft, S., Grimholt, U., Malmstrøm, M., Gregers, T.F., Rounge, T.B., Paulsen, J., Solbakken, M.H., Sharma, A., Wetten, O.F., Lanzén, A., Winer, R., Knight, J., Vogel, J.-H., Aken, B., Andersen, Ø., Lagesen, K., Tooming-Klunderud, A., Edvardsen, R.B., Tina, K.G., Espelund, M., Nepal, C., Previti, C., Karlsen, B.O., Moum, T., Skage, M., Berg, P.R., Gjøen, T., Kuhl, H., Thorsen, J., Malde, K., Reinhardt, R., Du, L., Johansen, S.D., Searle, S., Lien, S., Nilsen, F., Jonassen, I., Omholt, S.W., Stenseth, N.C., Jakobsen, K.S., 2011. The genome sequence of Atlantic cod reveals a unique immune system. Nature 477, 207–210.
Stockhammer, O.W., Zakrzewska, A., Hegedûs, Z., Spaink, H.P., Meijer, A.H., 2009. Transcriptome profiling and functional analyses of the zebrafish embryonic innate immune response to Salmonella Infection. J. Immunol. 182, 5641–5653.
Sudhagar, A., Kumar, G., El-Matbouli, M., 2018. Transcriptome analysis based on RNA-Seq in understanding pathogenic mechanisms of diseases and the immune system of fish: a comprehensive review. Int. J. Mol. Sci. 19, 245.
Sun, L., Liu, S., Wang, R., Jiang, Y., Zhang, Y., Zhang, J., Bao, L., Kaltenboeck, L., Dunham, R., Waldbieser, G., Liu, Z., 2014. Identification and analysis of genome-wide SNPs provide insight into signatures of selection and domestication in channel catfish (Ictalurus punctatus). PLoS One 9, e109666.
Sun, Y., Huang, Y., Li, X., Baldwin, C.C., Zhou, Z., Yan, Z., Crandall, K.A., Zhang, Y., Zhao, X., Wang, M., Wong, A., Fang, C., Zhang, X., Huang, H., Lopez, J.V., Kilfoyle, K., Zhang, Y., Ortí, G., Venkatesh, B., Shi, Q., 2016. Fish-T1K (Transcriptomes of 1,000 Fishes) Project: large-scale transcriptome data for fish evolution studies. GigaScience 5, 18.
Syahputra, K., Kania, P.W., Al-Jubury, A., Jafaar, R.M., Dirks, R.P., Buchmann, K., 2019. Transcriptomic analysis of immunity in rainbow trout (Oncorhynchus mykiss) gills infected by Ichthyophthirius multifiliis. Fish Shellfish Immunol. 86, 486–496.
Syaifudin, M., Bekaert, M., Taggart, J.B., Bartie, K.L., Wehner, S., Palaiokostas, C., Khan, M.G.Q., Selly, S.-L.C., Hulata, G., D'Cotta, H., Baroiller, J.-F., McAndrew, B.J., Penman, D.J., 2019. Species-specific marker discovery in Tilapia. Sci. Rep. 9, 13001.
Tan, S.C., Yiap, B.C., 2009. DNA, RNA, and protein extraction: the past and the present. J. Biomed. Biotechnol. 574398.
Tan, C.K., Natrah, I., Suyub, I.B., Edward, M.J., Kaman, N., Samsudin, A.A., 2019. Comparative study of gut microbiota in wild and captive Malaysian Mahseer (Tor tambroides). MicrobiologyOpen 8, e00734.
Tang, K.-Y., Wang, Z.-W., Wan, Q.-H., Fang, S.-G., 2019. Metagenomics reveals seasonal functional adaptation of the gut microbiome to host feeding and fasting in the Chinese alligator. Front. Microbiol. 10, 2409.
Tao, M., Zhou, Y., Li, S., Zhong, H., Hu, H., Yuan, L., Luo, M., Chen, J., Ren, L., Luo, J., Zhang, C., Liu, S., 2018. MicroRNA Alternations in the Testes Related to the Sterility of Triploid Fish. Mar. Biotechnol. 20, 739–749.
Terzibasi Tozzini, E., Savino, A., Ripa, R., Battistoni, G., Baumgart, M., Cellerino, A., 2014. Regulation of microRNA expression in the neuronal stem cell niches during aging of the short-lived annual fish Nothobranchius furzeri. Front. Cell. Neurosci. 8, 51.
Tian, C., Li, Z., Dong, Z., Huang, Y., Du, T., Chen, H., Jiang, D., Deng, S., Zhang, Y., Wanida, S., Shi, H., Wu, T., Zhu, C., Li, G., 2019. Transcriptome analysis of male and female mature gonads of silver sillago (Sillago sihama). Genes 10.
Tine, M., Kuhl, H., Gagnaire, P.-A., Louro, B., Desmarais, E., Martins, R.S.T., Hecht, J., Knaust, F., Belkhir, K., Klages, S., Dieterich, R., Stueber, K., Piferrer, F., Guinand, B., Bierne, N., Volckaert, F.A.M., Bargelloni, L., Power, D.M., Bonhomme, F., Canario, A. V.M., Reinhardt, R., 2014. European sea bass genome and its variation provide insights into adaptation to euryhalinity and speciation. Nat. Commun. 5, 5770.
Tsuchiya, C., Sakata, T., Sugita, H., 2008. Novel ecological niche of Cetobacterium somerae, an anaerobic bacterium in the intestinal tracts of freshwater fish. Lett. Appl. Microbiol. 46, 43–48.
Ulitsky, I., Shkumatava, A., Jan, C.H., Sive, H., Bartel, D.P., 2011. Conserved function of lincRNAs in vertebrate embryonic development despite rapid sequence evolution. Cell 147, 1537–1550.
Ulloa, P.E., Rincon, G., Islas-Trejo, A., Araneda, C., Iturra, P., Neira, R., Medrano, J.F., 2015. RNA sequencing to study gene expression and SNP variations associated with growth in zebrafish fed a plant protein-based diet. Mar. Biotechnol. 17, 353–363.
Vendrami, D.L.J., Houston, R.D., Gharbi, K., Telesca, L., Gutierrez, A.P., Gurney-Smith, H., Hasegawa, N., Boudry, P., Hoffman, J.I., 2019. Detailed insights into pan-European population structure and inbreeding in wild and hatchery Pacific oysters (*Crassostrea gigas*) revealed by genome-wide SNP data. Evol. Appl. 12, 519–534.
Venkatesh, B., Lee, A.P., Ravi, V., Maurya, A.K., Lian, M.M., Swann, J.B., Ohta, Y., Flajnik, M.F., Sutoh, Y., Kasahara, M., Hoon, S., Gangu, V., Roy, S.W., Irimia, M., Korzh, V., Kondrychyn, I., Lim, Z.W., Tay, B.-H., Tohari, S., Kong, K.W., Ho, S., Lorente-Galdos, B., Quilez, J., Marques-Bonet, T., Raney, B.J., Ingham, P.W., Tay, A., Hillier, L.W., Minx, P., Boehm, T., Wilson, R.K., Brenner, S., Warren, W.C., 2014. Elephant shark genome provides unique insights into gnathostome evolution. Nature 505, 174–179.
Vij, S., Kuhl, H., Kuznetsova, I.S., Komissarov, A., Yurchenko, A.A., Van Heusden, P., Singh, S., Thevasagayam, N.M., Prakki, S.R.S., Purushothaman, K., Saju, J.M., Jiang, J., Mbandi, S.K., Jonas, M., Tong, A. Hin Yan, Mwangi, S., Lau, D., Ngoh, S.Y., Liew, W.C., Shen, X., Hon, L.S., Drake, J.P., Boitano, M., Hall, R., Chin, C.-S., Lachumanan, R., Korlach, J., Trifonov, V., Kabilov, M., Tupikin, A., Green, D., Moxon, S., Garvin, T., Sedlazeck, F.J., Vurture, G.W., Gopalapillai, G., Katneni, V. Kumar, Noble, T.H., Scaria, V., Sivasubbu, S., Jerry, D.R., O'Brien, S.J., Schatz, M.C., Dalmay, T., Turner, S.W., Lok, S., Christoffels, A., Orbán, L., 2016. Chromosomal-level assembly of the asian seabass genome using long sequence reads and multi-layered scaffolding. PLoS Genet. 12, e1005954.
Wang, W., Yi, Q., Ma, L., Zhou, X., Zhao, H., Wang, X., Qi, J., Yu, H., Wang, Z., Zhang, Q., 2014. Sequencing and characterization of the transcriptome of half-smooth tongue sole (Cynoglossus semilaevis). BMC Genomics 15, 470.
Wang, Y., Lu, Y., Zhang, Y., Ning, Z., Li, Y., Zhao, Q., Lu, H., Huang, R., Xia, X., Feng, Q., Liang, X., Liu, K., Zhang, L., Lu, T., Huang, T., Fan, D., Weng, Q., Zhu, C., Lu, Y., Li, W., Wen, Z., Zhou, C., Tian, Q., Kang, X., Shi, M., Zhang, W., Jang, S., Du, F., He, S., Liao, L., Li, Y., Gui, B., He, H., Ning, Z., Yang, C., He, L., Luo, L., Yang, R., Luo, Q., Liu, X., Li, S., Huang, W., Xiao, L., Lin, H., Han, B., Zhu, Z., 2015. The draft genome of the grass carp (Ctenopharyngodon idellus) provides insights into its evolution and vegetarian adaptation. Nat. Genet. 47, 625–631.

Wang, S.Y., Lau, K., Lai, K.-P., Zhang, J.-W., Tse, A.C.-K., Li, J.-W., Tong, Y., Chan, T.-F., Wong, C.K.-C., Chiu, J.M.-Y., Au, D.W.-T., Wong, A.S.-T., Kong, R.Y.-C., Wu, R.S.-S., 2016. Hypoxia causes transgenerational impairments in reproduction of fish. Nat. Commun. 7, 12114.

Wang, Y., Zhang, M., Qin, Q., Peng, Y., Huang, X., Wang, C., Cao, L., Li, W., Tao, M., Zhang, C., Liu, S., 2019. Transcriptome profile analysis on ovarian tissues of autotetraploid fish and diploid Red Crucian Carp. Front. Genet. 10, 208.

Watanabe, Y., Kinoshita, A., Yamada, T., Ohta, T., Kishino, T., Matsumoto, N., Ishikawa, M., Niikawa, N., Yoshiura, K., 2002. A catalog of 106 single-nucleotide polymorphisms (SNPs) and 11 other types of variations in genes for transforming growth factor-beta1 (TGF-beta1) and its signaling pathway. J. Hum. Genet. 47, 478–483.

Wong, S., Waldrop, T., Summerfelt, S., Davidson, J., Barrows, F., Kenney, P.B., Welch, T., Wiens, G.D., Snekvik, K., Rawls, J.F., Good, C., 2013. Aquacultured Rainbow Trout (Oncorhynchus mykiss) Possess a Large Core Intestinal Microbiota That Is Resistant to Variation in Diet and Rearing Density. Appl. Environ. Microbiol. 79, 4974–4984.

Wu, C., Zhang, D., Kan, M., Lv, Z., Zhu, A., Su, Y., Zhou, D., Zhang, J., Zhang, Z., Xu, M., Jiang, L., Guo, B., Wang, T., Chi, C., Mao, Y., Zhou, J., Yu, X., Wang, H., Weng, X., Jin, J.G., Ye, J., He, L., Liu, Y., 2014. The draft genome of the large yellow croaker reveals well-developed innate immunity. Nat. Commun. 5, 5227.

Xie, M., Ming, Y., Shao, F., Jian, J., Zhang, Y., Peng, Z., 2018. Restriction site-associated DNA sequencing for SNP discovery and high-density genetic map construction in southern catfish (Silurus meridionalis), 5, 172054.

Xu, J., Ji, P., Zhao, Z., Zhang, Y., Feng, J., Wang, J., Li, J., Zhang, X., Zhao, L., Liu, G., Xu, P., Sun, X., 2012. Genome-wide SNP discovery from transcriptome of four common carp strains. PLoS One 7, e48140.

Xu, P., Zhang, X., Wang, X., Li, J., Liu, G., Kuang, Y., Xu, J., Zheng, X., Ren, L., Wang, G., Zhang, Y., Huo, L., Zhao, Z., Cao, D., Lu, C., Li, C., Zhou, Y., Liu, Z., Fan, Z., Shan, G., Li, X., Wu, S., Song, L., Hou, G., Jiang, Y., Jeney, Z., Yu, D., Wang, L., Shao, C., Song, L., Sun, J., Ji, P., Wang, J., Li, Q., Xu, L., Sun, F., Feng, J., Wang, C., Wang, S., Wang, B., Li, Y., Zhu, Y., Xue, W., Zhao, L., Wang, J., Gu, Y., Lv, W., Wu, K., Xiao, J., Wu, J., Zhang, Z., Yu, J., Sun, X., 2014a. Genome sequence and genetic diversity of the common carp, Cyprinus carpio. Nat. Genet. 46, 1212–1219.

Xu, Y., Guan, L., Shen, T., Zhang, J., Xiao, X., Jiang, H., Li, S., Yang, J., Jia, X., Yin, Y., Guo, X., Wang, J., Zhang, Q., 2014b. Mutations of 60 known causative genes in 157 families with retinitis pigmentosa based on exome sequencing. Hum. Genet. 133, 1255–1271.

Xu, H., Dong, X., Zhang, Z., Yang, M., Wu, X., Liu, H., Lao, Q., Li, C., 2015. Assessment of immunotoxicity of dibutyl phthalate using live zebrafish embryos. Fish Shellfish Immunol. 45, 286–292.

Yadetie, F., Zhang, X., Hanna, E.M., Aranguren-Abadia, L., Eide, M., Blaser, N., Brun, M., Jonassen, I., Goksoyr, A., Karlsen, O.A., 2018. RNA-Seq analysis of transcriptome responses in Atlantic cod (Gadus morhua) precision-cut liver slices exposed to benzo [a]pyrene and 17alpha-ethynylestradiol. Aquat. Toxicol. 201, 174–186.

Yan, B., Liu, B., Zhu, C.-D., Li, K.-L., Yue, L.-J., Zhao, J.-L., Gong, X.-L., Wang, C.-H., 2013. microRNA regulation of skin pigmentation in fish. J. Cell Sci. 126, 3401–3408.

Yang, W., Chen, H., Cui, X., Zhang, K., Jiang, D., Deng, S., Zhu, C., Li, G., 2018. Sequencing, de novo assembly and characterization of the spotted scat Scatophagus argus (Linnaeus 1766) transcriptome for discovery of reproduction related genes and SSRs. J. Oceanol. Limnol. 36, 1329–1341.

Ye, L., Amberg, J., Chapman, D., Gaikowski, M., Liu, W.-T., 2014. Fish gut microbiota analysis differentiates physiology and behavior of invasive Asian carp and indigenous American fish. ISME J. 8, 541–551.

Yu, Y., Wei, J., Zhang, X., Liu, J., Liu, C., Li, F., Xiang, J., 2014. SNP discovery in the transcriptome of white Pacific shrimp Litopenaeus vannamei by next generation sequencing. PLoS One 9, e87218.

Zampiga, M., Flees, J., Meluzzi, A., Dridi, S., Sirri, F., 2018. Application of omics technologies for a deeper insight into quali-quantitative production traits in broiler chickens: A review. J. Anim. Sci. Biotechnol. 9 (1), 1–18.

Zarkasi, K.Z., Taylor, R.S., Abell, G.C.J., Tamplin, M.L., Glencross, B.D., Bowman, J.P., 2016. Atlantic Salmon (Salmo salar L.) Gastrointestinal Microbial Community Dynamics in Relation to Digesta Properties and Diet. Microb. Ecol. 71, 589–603.

Zhang, Y.-X., Barry, J., Moore, D., Amitay, S., Zhang, et al., 2012. PLoS One 2013.

Zhang, X., Wen, H., Wang, H., Ren, Y., Zhao, J., Li, Y., 2017a. RNA-Seq analysis of salinity stress–responsive transcriptome in the liver of spotted sea bass (Lateolabrax maculatus). PLoS One 12, e0173238.

Zhang, G.-Q., Liu, B., Li, J., Luo, C.-Q., Zhang, Q., Chen, J.-L., Sinha, A., Li, Z.-Y., 2017b. Fish intake during pregnancy or infancy and allergic outcomes in children: A systematic review and meta-analysis. Pediatr. Allergy Immunol. 28, 152–161.

Zhang, Z., Yu, A., Lan, J., Zhang, Y., Zhang, H., Li, Y., Hu, M., Cheng, J., Wei, S., Lin, L., 2017c. Draft Genome Sequence of an Attenuated Streptococcus agalactiae Strain Isolated from the Gut of a Nile Tilapia (Oreochromis niloticus). Genome Announc. 5 (e01627-01616).

Zhang, Y., Qin, C., Yang, L., Lu, R., Zhao, X., Nie, G., 2018. A comparative genomics study of carbohydrate/glucose metabolic genes: from fish to mammals. BMC Genomics 19, 246.

Zhang, H., Wang, Q., Liu, S., Huo, D., Zhao, J., Zhang, L., Zhao, Y., Sun, L., Yang, H., 2019. Genomic and metagenomic insights into the microbial community in the regenerating intestine of the sea cucumber Apostichopus japonicus. Front. Microbiol. 10, 1165.

Zheng, H.-T., Gong, Y., Chen, F., Liang, Y., Huang, D., 2018. Length-weight relationships for three fishes in the Mali Hka River and Nmai Hka River, Myanmar. J. Appl. Ichthyol. 34, 1228–1229.

Zhu, H., Leung, S.W., 2015. Identification of microRNA biomarkers in type 2 diabetes: a meta-analysis of controlled profiling studies. Diabetologia 58, 900–911.